Week 9: International Bonds, Equities, and Portfolio Choice

FINA3020: International Finance

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Fall 2026

Review: Calculate Return on Foreign-Currency Investment

A Hong Kong investor buys a European bond for EUR 360. The spot rate is 9 HKD per EUR. One year later, the bond repays EUR 360 plus EUR 40 interest. The new spot rate is 8 HKD per EUR. The investor has no currency hedge. Ignore transaction costs and taxes.

1. Calculate the profit or loss in HKD. Then calculate the percentage return.

2. The initial one-year forward rate was 8.1 HKD per EUR. Assume covered interest parity (CIP) holds. What does this imply about HKD and EUR interest rates?

1) Purchase EUR 360 at spot rate 9 HKD per EUR. Initial expenditure is $360 \times 9 = 3240$ HKD

One year later, you have EUR 400 in total (including EUR 40 of interest). At the new spot rate of 8 HKD per EUR, this is $400 \times 8 = 3200$ HKD.

You have a loss of 40 HKD, with percentage loss of $\frac{40}{3240} \approx 1.2\%$

2) Under CIP, $F/S = (1 + i_{\mathrm{HKD}})/(1 + i_{\mathrm{EUR}}) = 0.9$. Thus, $1 + i_{\mathrm{HKD}}$ is 90% of $1 + i_{\mathrm{EUR}}$. For example, HKD interest of 8% and EUR interest of 20% satisfy $1.08/1.20 = 0.9$.

A Bond: Funding Today and Payments Later [Core Concept]

A **bond** is a tradable debt claim. The **issuer** borrows from investors when it sells a new bond (primary market). The investor becomes a creditor.

Contract term	Meaning in a simple fixed-rate bond
Principal / face value / par	Amount the issuer promises to repay at the final date.
Maturity	Date when that principal becomes due.
Coupon	Scheduled interest payment. Coupon rate is usually percent of face value.
Issue proceeds	Cash that the issuer receives, after issue costs.

Example: A bond has USD 100 face value, a 5% annual coupon, and two years to maturity. If the issuer pays in full, the investor receives USD 5 in Year 1 and USD 105 in Year 2.

A bond's price can differ from its face value. Its **yield to maturity** is the discount rate that makes the promised payments' present value equal to that price.

Issuer, Currency, and Market

[Core Concept]

Term	Example	What the label describes
Domestic bond	Japanese firm, JPY, Japan	Home issuer, currency, and market
Foreign bond	Japanese firm, USD, US	Foreign issuer in a domestic market
Eurobond	Japanese firm, USD, offshore	Issue outside the currency's home market

- A USD foreign bond in the US is a Yankee bond. The "Euro" in Eurobond refers to the market structure, and the currency can be USD.
- A guarantor promises to meet specified payments if the issuer cannot.

For an HKD investor, which column tells us the currency of the promised payments?

Historical debt by currency →

Toyota Motor Corporation: A USD Bond Issue

[Background Context]

In June 2025, Toyota Motor Corporation issued three USD 500M sustainability bonds. The legal issuer was the Japanese parent company.

Maturity date	Principal	Annual coupon	Payment frequency
30 June 2027	USD 500M	4.186%	Twice a year
30 June 2030	USD 500M	4.450%	Twice a year
30 June 2035	USD 500M	5.053%	Twice a year

For USD 2,000 face value of the 2030 bond, calculate one full half-year coupon and the final scheduled payment. Assume no early repayment or default.

Source: Toyota Motor Corporation, June 2025 prospectus supplement. Minimum face-value purchase USD 2,000.

Example of fixed bond factsheet →

Toyota tranche calculation →

Toyota: The Promised Cash Flows

[Background Context]

For USD 2,000 face value of the 4.450% notes:

$$\text{Half-year coupon} = 2{,}000 \times 0.04450/2 = \textbf{USD 44.50}.$$

$$\text{Final scheduled payment} = 2{,}000 + 44.50 = \textbf{USD 2,044.50}.$$

If on 1 January 2030, you sold USD 2,000 face value for USD 1,900, the yield to maturity would be $\left(\frac{2044.50}{1900} - 1\right) \times 2 = 15.2\%$

- Coupons are paid on 30 June and 30 December, with final payment on 30 June 2030.
- Toyota assigns an amount equal to the net proceeds to eligible projects. Its Woven Planet Bond Framework defines the environmental and social project categories.

Source: Toyota, 2025 prospectus supplement.

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Source: Toyota, 2025 prospectus supplement.

Fixed and Floating Payments

[Core Concept]

The principal is USD 100M, with annual payments. The fixed coupon rate is 5%. The floating rate equals a reference rate, or **benchmark**, plus 150 basis points. This spread equals 1.50 percentage points.

Benchmark at reset	Fixed coupon	Floating coupon
3.5%	USD 5M	$100(0.035 + 0.015) = \text{USD } 5\text{M}$
5.5%	USD 5M	$100(0.055 + 0.015) = \text{USD } 7\text{M}$

- If the discount rate rises, the price of a fixed-rate bond falls.
- A floating-rate reset changes the borrower's interest payment. This reduces the bond price's sensitivity to benchmark changes.

Duration mechanics →

An Interest-Rate Swap Changes the Interest Exposure

[Core Concept]

An **interest-rate swap (IRS)** exchanges interest payments. For example, in a fixed-for-floating swap, one party pays a fixed rate and receives a reference rate from the other party.

- The **notional** is the amount used to calculate the payments. In this standard same-currency example, the parties do not lend or exchange that principal.
- Suppose a firm owes USD 100M at benchmark $L + 1.5$ percentage points. It separately agrees to pay 3.5% fixed and receive L on a USD 100M swap.
- Each year, the firm pays the lender $100(L + 1.5\%)$ million dollars. Separately, it pays the swap counterparty USD 3.5M and receives $100L$ million dollars. The **counterparty** is the other party to the swap.
- The swap's floating receipt offsets the benchmark part of the loan payment. The firm still pays the loan spread and the swap's fixed rate.

Calculate the firm's combined annual interest cost when $L = 3.5\%$ and when $L = 5.5\%$.

Source: BIS, interest-rate swap definition.

The Loan and Swap Have Separate Cash Flows

[Background Context]

All amounts are USD millions for one year on USD 100M. The firm pays its lender and exchanges the swap cash flows with a separate counterparty.

Firm's cash flow	$L = 3.5\%$	$L = 5.5\%$
Pay loan interest, $100(L + 1.5\%)$	5.0	7.0
Pay fixed interest on swap	3.5	3.5
Receive floating interest on swap	-3.5	-5.5
Combined interest cost	5.0	5.0

Net rate: $(L + 1.5\%) + 3.5\% - L = 5\%$.

- The firm still owes the loan principal. The swap changes its net interest exposure.
- A **cross-currency swap** exchanges payments in different currencies and can exchange principal.

The Swap Rate and the Borrower's Funding Rate

- For a fixed-for-floating interest-rate swap, the quoted **swap rate** is usually its fixed rate. At the start, the par swap rate makes the two payment streams equal in present value, before dealer adjustments.
- The floating payment follows the agreed benchmark and reset rule. USD swaps often use SOFR. The quote depends on market rates, maturity, and payment rules.
- The firm's **borrowing rate** belongs to its loan or bond. In our example, the firm pays 3.5% fixed and receives the loan's benchmark through the swap. Its combined interest cost is $3.5\% + 1.5\% = 5\%$.
- Dealer charges and collateral funding can add costs.

The contract in this example uses 3.5%.

Source: CME Group, pricing and hedging USD SOFR interest-rate swaps.

A Fixed-to-Floating Bond

[Core Concept]

HSBC Holdings issued USD 2.25B of notes in March 2023, maturing on 9 March 2034. The contract has two interest-rate periods.

Period	Interest terms
Before 9 March 2033	6.254% fixed annual rate, with half-year payments.
From 9 March 2033	Compounded daily SOFR plus 2.390 percentage points, paid quarterly.

- **SOFR**, the Secured Overnight Financing Rate, measures secured overnight USD borrowing. Compounding adds interest on interest that has already accumulated.
- A **reset** updates the rate under the contract. The margin is 239 basis points. An **optional redemption date** permits the issuer to repay early under specified terms.

Source: HSBC, registered-security terms, US404280DV88.

Coupon Rate, Market Price, and Yield

[Core Concept]

A two-year bond has face value 100 and annual coupon 5. It pays 5 in Year 1 and 105 in Year 2.

- **Coupon rate**: annual interest divided by face value, here $5/100 = 5\%$.
- **Market price** P: what a buyer pays today. A price of 98 means 98% of face value when quoted per 100.
- **Yield to maturity** y: the annual discount rate that makes the promised payments' present value equal to today's price.

$$P = \frac{5}{1+y} + \frac{105}{(1+y)^2}.$$

A bond trades at a **discount** when its price is below face value. It trades at a **premium** when its price is above face value. For unchanged promised payments, a higher yield means a lower price.

Source: SEC Investor.gov, What Are Corporate Bonds?.

Bond Pricing: The Investor Needs a Sale Value

An investor owns a bond with face value 100, annual coupon 5, and two years remaining. Assume no default risk.

Date	Coupon	Principal
One year from today	5	0
Two years from today	5	100

If the yield is 6%, what is the price?

Bond Price: Discount Each Payment Separately [Core Concept]

Face value 100, annual coupon 5, two years to maturity, annual yield 6%. Assume no default risk.

1. Write the cash flows: Year 1 pays 5. Year 2 pays coupon 5 plus principal 100, or 105.
2. Discount Year 1: $5/1.06 = 4.71698$.
3. Discount Year 2: $105/1.06^2 = 93.44963$.
4. Add them: $P = 4.71698 + 93.44963 = 98.16661 \approx 98.17$.

The coupon rate, 5%, is below the required yield, 6%, so the bond sells below par.

A Fixed-Rate Bond: Reading the Contract Fields

The issuer is CK Hutchison International (20) Limited

Field	Reported term and interpretation
Issue / maturity	8 May 2020 / 8 May 2050: a 30-year contract at issue.
Currency / issue size	USD / USD 750M total face value.
Annual coupon	3.375%, paid twice a year.
Issue price / yield	99.141 per 100 face / 3.421% annual yield.
Senior unsecured	Ranks ahead of lower-priority debt, with no specific pledged asset.

For USD 1,000 face value, each full half-year coupon is $1{,}000(0.03375)/2 = 16.875$. Its issue yield exceeds its coupon rate because the issue price is below par.

Source: BondSuperMart factsheet, USG2181LAB92.

← Back to Main Slide

Credit and Currency Risk

[Core Concept]

- **Credit risk**: the issuer may fail to pay. A credit rating assesses repayment ability. The credit spread is the yield difference from a comparable benchmark bond.
- **Issuer currency mismatch**: the issuer receives income in one currency and owes payments in another.
- **Investor currency exposure**: the bond pays in a currency different from the investor's reporting currency.
- **Market price**: higher yields reduce the price of fixed payments. This can happen when benchmark rates rise or credit spreads increase. Some investors must record these price losses before they sell.

Sovereign spillovers →

Rating categories → Historical defaults →

A Treasury Decision: Funding a USD Project

[Background Context]

A multinational company needs USD 100M today for equipment serving US customers. Expected operating receipts are in USD. It compares five-year funding with principal due in Year 5.

Contract	Terms used in this exercise
Fixed bond	5.00% annual coupon
Floating bond	Annual benchmark + 1.50 percentage points
Floating loan	Benchmark + 1.00 point, with the reset rule below

- Each raises USD 100M net today. Ignore other fees, taxes, collateral, and default.
- At each reset after Year 1, the lender chooses between the previous loan rate and benchmark + 1.00 percentage point. Assume it chooses the higher rate. The contract allows the lender to reject rate cuts.

PV solution →

Three Funding Choices for USD 100M

A firm borrows USD 100M for five years and repays the principal in Year 5. Here is an example of what the **realized** rates look like. The firm's initial decision depends on **expected** rates instead

Year	Benchmark	Fixed bond	Floating bond	Floating loan
1	3.0%	5.0	4.5	4.0
2	4.5%	5.0	6.0	5.5
3	3.5%	5.0	5.0	5.5
4	5.0%	5.0	6.5	6.0
5	4.0%	5.0	5.5	6.0

Present Value of the Financing Costs

[Core Concept]

Discount the previous slide's payments at a common 5.2% rate. The calculation compares costs for one specified interest-rate path.

$$PV = \sum_{t=1}^{5} \frac{C_t}{1.052^t} + \frac{100}{1.052^5}.$$

C_t is the interest payment in Year t, per USD 100 principal.

Instrument	PV per USD 100 raised
Fixed bond	99.14
Floating bond	101.18
Floating loan	100.66

- The fixed bond has the lowest PV under this assumed rate path.
- Compare a falling-rate path and the cash needed in weak business conditions.

Payments → PV Table → Business States →

Funding Costs under the Falling-Rate Path

Now suppose the **realized** annual benchmark were 3.5%, 3.0%, 2.5%, 2.0%, and 1.5%. The loan starts at 4.5% and follows the same lender reset rule. All amounts are USD millions. Add the common 100M principal payment in Year 5.

Contract	Year 1	2	3	4	5	Total
Fixed bond	5	5	5	5	5	25.00
Floating bond	5	4.5	4	3.5	3	20.00
Floating loan	4.5	4.5	4.5	4.5	4.5	22.50

- The floating bond now has the lowest interest total, USD 20M. The lender retains the loan's initial 4.5% rate, so the loan costs USD 22.5M.
- Before borrowing, the future path is uncertain. The firm must consider how interest payments behave when its operating cash receipts are weak.

Funding Choice under Weak Business Conditions

The face value is USD 100M. The fixed bond pays 5%. The floating bond pays the benchmark plus 1.50 percentage points.

Scenario	Cash before interest	Benchmark
Demand recession	USD 6M	2.0%
Input cost shock with high rates	USD 6M	5.5%

1. Calculate cash remaining after fixed and floating interest in each scenario.

Financing Costs Depend on the Business State

[Core Concept]

Scenario	Fixed interest	Float interest	Cash after fixed	Cash after float
Demand recession	5.0	3.5	1.0	2.5
Cost shock	5.0	7.0	1.0	−1.0

- Amounts are USD millions. Floating debt preserves cash when the benchmark falls with sales. A rate rise during weak cash flow instead creates a USD 1M funding gap.

A **funding gap** means the firm cannot pay this year's interest from operating cash. It must use reserves, obtain new funding, sell assets, or miss a payment.

Floating debt helps when weak demand lowers both sales and interest rates. Fixed debt is more useful when input costs reduce cash flow while inflation keeps interest rates high.

Additional business states → Further funding exercise →

A Financing Decision across Business States

The table shows one-year outcomes for USD 100M of debt. Principal is due after this year. If the firm cannot pay USD 1M of interest, assume it incurs an additional USD 4M disruption cost.

State	Probability	Cash before interest	Fixed	Floating
Expansion	50%	12	5	6
Demand recession	30%	6	5	3.5
Input cost shock	20%	6	5	7

1. Calculate expected interest and the probability of a funding gap for each bond.
2. Include the disruption cost. Which bond has the lower expected total cost?

Expected Financing Cost and the Funding Gap [Core Concept]

The probabilities are 50% expansion, 30% demand recession, and 20% input cost shock.

USD millions	Fixed bond	Floating bond
Expected interest	5.00	$0.50(6) + 0.30(3.5) + 0.20(7) = 5.45$
Probability of a funding gap	0%	20%
Expected additional disruption cost	0	$0.20(4) = 0.80$
Expected interest + disruption	5.00	6.25

- The fixed bond has lower expected total cost in this model. The floating bond preserves more cash specifically in the demand recession.
- A reserve or committed credit line could avoid disruption. Compare its cost and availability with changing the interest-rate exposure.

BP: Committed Credit during the 2020 Funding Shock

[Background Context]

BP is an energy company. In March 2020, it obtained a USD 10B revolving credit facility. In April, it issued USD 6.8B of bonds. Its annual report records these amounts.

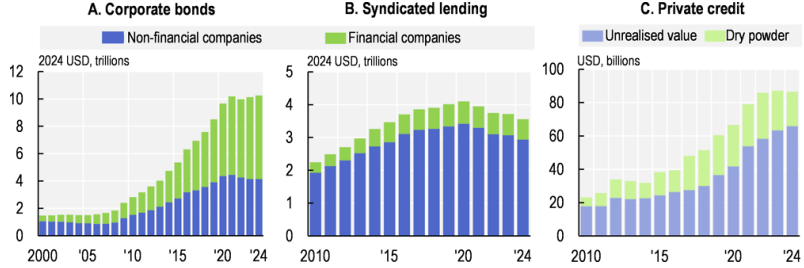
- The pandemic disrupted business cash flows and funding. A committed credit line gave BP access to borrowing capacity under agreed conditions.
- If BP borrows USD 1B from the facility, cash and debt each rise by USD 1B. Unused borrowing capacity falls to USD 9B.

Interpretation: Why might a firm get a credit line and issue bonds during the same period?

Source: BP, Annual Report 2020.

Corporate Funding in Asia: Reading the Evidence

[Background Context]



- **Syndicated loan**: several lenders share one loan. **Private credit**: lending arranged outside public bond markets, often by investment funds.
- **Unrealised value** is the reported value of existing fund investments. **Dry powder** is committed investor capital still available to invest.

Panels A/B use USD trillions and C uses billions. What bank-credit data are missing from these panels?

Source: OECD, Asia Capital Markets Report 2025. Historical figures through 2024.

Sovereign debt chart → | Corporate debt chart → | Asian bond markets → | Asian syndicated loans →

Equity: Ownership and the Residual Claim [Core Concept]

Equity is ownership in a company. A **share** is one unit of ownership.

- Creditors have contractual claims. Common shareholders have a **residual claim**. They receive the value left after the company meets its other obligations.
- A **dividend** distributes cash to shareholders. A company can also return cash by buying back shares. The company decides whether to make these payments.
- Shareholders can gain or lose when the market price changes. If a company fails, common shareholders can lose their entire investment.

Debt and equity provide funding with different payment flow and risk profiles for investors.

Who Receives the Money in an Equity Trade?

[Core Concept]

Transaction	Buyer receives	Cash recipient
New share issue	Newly issued ownership claim	Issuing company
Secondary share trade	Existing ownership claim	Selling investor

- A primary market sells newly issued claims and raises issuer funds. A secondary market trades existing claims between investors.
- Cross-listing means making a company's shares available on another exchange. It can widen investor access and add disclosure obligations. Buying an existing cross-listed share transfers an existing claim.

Check: A Hong Kong investor buys a Japanese share from another investor. Does this trade raise new cash for the company?

Market institutions →

An Initial Public Offering

[Core Concept]

An **initial public offering (IPO)** sells shares to public investors as a firm enters the public market.

1. The firm chooses how much capital to raise and appoints advisers. In Hong Kong, a sponsor checks the firm's information and supports its listing application.
2. The firm prepares audited accounts and a prospectus explaining its business, risks, and use of funds. The exchange reviews listing eligibility.
3. Investment banks market the shares and collect investor orders. This **bookbuilding** helps choose the offer price and allocations.
4. Investors pay for the offer shares. New-share proceeds go to the firm, less fees. Proceeds from existing shares sold in the offer go to those existing shareholders.

Source: HKEX.

Investment Banks and the Offer Price

[Background Context]

A firm sells 20M new shares at HKD 10 each. Assume underwriting and other issue costs total 4% of gross proceeds.

$$\text{Gross proceeds} = 20\text{M}(10) = \text{HKD } 200\text{M}, \qquad \text{Net proceeds} = 200(0.96) = 192\text{M}.$$

- Banks introduce institutional investors and collect demand at different prices. An offer price must balance proceeds against the risk of an unsuccessful sale.
- Under a **firm commitment**, the bank agrees to buy the issue under the contract's terms. It bears the risk of unsold shares. Other agreements require only an effort to sell the shares.
- If the secondary price rises to HKD 12, the original issue still raised HKD 200M gross. Later buyers pay existing holders.

Choosing a Listing Venue

[Background Context]

Consider a mainland consumer firm that wants to raise equity from investors outside the mainland.

Consideration	How it affects the choice
Investor demand	Hong Kong may offer investors familiar with its regional business. Investing preferences differ by location.
Access and liquidity	Compare the investors who can buy and likely trading after listing, including Stock Connect eligibility.
Rules and costs	Compare listing eligibility, disclosure, governance, legal exposure, and ongoing compliance costs.
Currency and operations	Trading in HKD, CNH, or USD changes settlement currency, separately from the firm's business exposure.

Source: HKEX, listing overview.

A listed firm may issue additional shares to finance expansion, acquire another business, or repay debt. This is a **follow-on offering**.

- A **placing** sells shares to selected investors. A **rights issue** gives existing shareholders the right to buy new shares on stated terms.
- New shares increase the total share count. If this rises from 100M to 125M, an unchanged holding of 1M shares falls from 1% ownership to 0.8%.
- The firm receives cash for the new shares. The issue price and use of this cash affect each share's value.
- When an existing shareholder sells shares, that shareholder receives the cash.

A Share Return Includes the Cash Distribution

An investor buys one share for EUR 100. At year-end, the investor receives a EUR 3 dividend and sells the share for EUR 105. Ignore tax and trading costs.

1. Calculate the price gain in euros and as a percentage of the purchase price.
2. Calculate the dividend yield, defined here as the cash dividend divided by the initial purchase price.
3. Calculate the total return using both the sale proceeds and the dividend. Does a 5% price rise describe the full investment outcome?

Price Gain, Dividend Yield, and Total Return

For the same EUR 100 purchase:

Component	Calculation	Return
Price gain	$(105 - 100)/100$	5%
Dividend yield	$3/100$	3%
Total return	$(105 + 3 - 100)/100$	8%

- **Capital gain** is the increase in the share's price. Total return also includes cash paid to the investor.
- For a dividend paid at year-end, $R = (P_1 + D - P_0)/P_0$. Here P_0 is the purchase price, P_1 is the sale price, and D is the dividend.

Why Use an American Depositary Receipt (ADR)?

[Background Context]

A US investor wants exposure to a foreign company but normally trades through a US broker.

- An ADR represents foreign shares and trades in the US market in USD. A depositary bank arranges safekeeping of the shares and converts dividends.
- The investor still bears the foreign business's risks and currency exposure. Depositary fees and conversion terms also affect the return.
- The ADR terms state how many shares each receipt represents. To profit from a price difference, a trader must be able to convert between them, incurring transaction costs and sometimes facing restrictions on flows.

Source: SEC Investor.gov, American Depositary Receipts.

American Depositary Receipts (ADRs)

[Background Context]

A depositary bank holds or arranges custody of foreign shares and issues US-traded receipts representing them. These receipts are American depositary receipts (ADRs). Assume one ADR represents **two Hong Kong shares**, each worth HKD 80. Spot is **7.80 HKD per USD**.

- The underlying package costs $2 \times 80 =$ HKD 160.
- Express the same package in USD by dividing by 7.80 HKD/USD. Keep the units on the calculation.
- Assume a trader can acquire the shares, convert them into an ADR, and sell the ADR at USD 21.00. All costs total USD 0.20 per ADR.

Another conversion example →

ADRs: A Price Gap after Trading Costs

$$P^*_{\mathrm{ADR}} = \frac{2 \times 80 \text{ HKD}}{7.80 \text{ HKD/USD}} = \text{USD } 20.5128.$$

$$\text{Net margin} = 21.00 - 20.5128 - 0.20 = \textbf{USD 0.2872 per ADR}.$$

1. Acquire two shares, arrange depositary conversion, and sell the receipt. Funding and the sale must cover the time between those steps.
2. Market hours, conversion delays, and price changes can erase the quoted margin. A short sale sells a borrowed receipt, so borrowing access is required.
3. With an ADR sale price of USD 20.60, the same trade loses about USD 0.1128 after costs. Trading costs can exceed a small price difference.

Stock Connect: Access through Brokers and Clearing [Core Concept]

- **Stock Connect** links Hong Kong trading access with eligible shares in Shanghai and Shenzhen. A **broker** submits an investor's order to the market.
- A **northbound** order buys eligible mainland shares through Hong Kong. A **southbound** order buys eligible Hong Kong shares through the mainland connection.
- **Clearing** determines payment and delivery obligations. **Settlement** transfers cash and securities. HKSCC and ChinaClear support these processes.
- **A-shares** trade in mainland China in RMB. **H-shares** are shares of mainland-incorporated firms listed in Hong Kong. Investors generally cannot convert a company's A-shares into its H-shares, or the reverse.

Why can an ADR conversion trade be possible when the same trade between A-shares and H-shares is unavailable?

Institutional details →

Historical Connect flows →

Equity Exposure through a Total-Return Swap [Core Concept]

A fund uses a swap to receive the total return on a specified foreign share or index. It pays financing charges to the dealer.

- The **reference notional** is the amount used to calculate swap payments. The fund usually provides collateral. It does not pay the full share purchase price.
- The equity payment includes price changes and the dividends specified in the contract. A negative total return creates a payment from the fund to the dealer.
- The financing payment usually uses a reference rate plus an agreed spread. USD contracts often use SOFR. The contract defines reset dates, interest-day counting, and the calculation amount.
- The fund has a claim under the swap contract. It receives no shareholder voting rights through the swap itself.

Source: CME Group, equity index swap mechanics.

The Swap Financing Rate and the Cash Payment

[Core Concept]

The swap notional is USD 10M. Interest uses 90 actual days divided by 360 (ACT/360). The annual benchmark is 4% and the spread is 2 percentage points.

$$\text{Financing payment} = N(b+s)\Delta = 10\text{M}(4\% + 2\%)\frac{90}{360} = \textbf{USD 150,000}.$$

Period's total equity return	$+8\%$	-8%
Equity leg received by fund	800,000	$-800{,}000$
Financing leg paid by fund	150,000	150,000
Net swap cash flow to fund	650,000	$-950{,}000$

The 6% rate is annual. Here N is notional, b the benchmark, s the spread, and Δ the year fraction. Ignore collateral interest, fees, taxes, and margin transfers during the period.

The Dealer's Hedge, Funding, and Collateral

Suppose the dealer promises a fund the return on USD 10M of shares.

- One hedge is to buy the shares. The dealer then needs USD 10M of purchase funding, arranged through its own cash and borrowing. The shares' return offsets what the dealer owes the fund on the equity part of the swap.
- A dealer can also offset exposure against other clients, facing non-payment risk, or use derivatives
- The financing spread covers funding, capital, trading, and hedging costs. It also compensates the dealer for counterparty risk.
- Client collateral protects the dealer if the client fails to pay. The agreement and rules determine whether the dealer can reuse it. Some collateral must remain separate.

Source: Hedge cash flows: illustrative replication. SEC, capital, margin, segregation, and risk controls.

Requirements on Swap Dealers

The applicable rules depend on the legal entity, jurisdiction, product, client, and clearing arrangement.

Requirement	Purpose and implementation
Authorization	Obtain permissions. Check client eligibility. Explain risks and control conflicts.
Capital and controls	Hold capital for losses. Set risk limits, value positions, and plan cash needs.
Margin	Variation margin covers current exposure. Initial margin covers possible losses during closure after default, where required.
Operations	Confirm and report trades. Compare records. Manage disputes and collateral.

Dealer capital absorbs losses. Contract margin covers counterparty exposure. Required amounts depend on the applicable rules and contract.

US and Hong Kong examples →

Source: US Exchange Act, Section 15F; HKMA CR-G-14.

Equity Perpetual Futures

[Core Concept]

An **equity perpetual future**, or **perp**, follows a share, ETF, or equity index. It usually has no scheduled expiry date. The trading platform sets its contract rules.

- The trader provides margin. A long position gains when the price rises. A short position gains when the price falls. The contract specifies payment currency and position size.
- Periodic **funding payments** help keep the perp price near its reference index. The payment direction can change while the position remains open.
- A perp gives no ownership or vote in the underlying company. Dividend and corporate-action adjustments depend on the venue's rules.
- Some equity perps trade 24 hours a day. When the stock exchange is closed, hedging and estimating current share value become harder.

Perp Funding: Who Pays Whom? [Core Concept]

- Under the usual rule, a **positive funding rate** means holders of long positions pay holders of short positions. A negative rate reverses this payment.
- When the perp price exceeds its reference price, the funding rate generally rises. Each platform defines the calculation and any limits.
- The platform calculates funding using the contract's **position notional**, the value of the position. The margin deposit can be smaller. Trading and liquidation fees are separate.

For a linear contract with one payment at each interval end:

$$\text{Long's funding cost at } t = qM_t f_t,$$

Here q is the number of units. M_t is the platform's funding price in USD per unit. f_t is the interval's funding rate, including its sign.

A rate of 0.01% per 8 hours is $0.01\% \times 3 \times 365 = 10.95\%$ on a simple annualized basis *if maintained*. The next interval's rate can change.

Exercise: A Foreign Equity Perp and Its Funding Costs

[Background Context]

A Hong Kong investor buys 100 units of a USD equity perp at USD 100 each. Initial cash collateral is USD 2,000. A USD 1 price change gives a USD 1 profit or loss per unit.

8-hour interval ending	Funding price (USD)	Interval funding rate
1	100	+0.01%
2	102	+0.02%
3	101	−0.01%

The investor keeps the position through all three payment times, then closes it at USD 104. Ignore fees, dividends, collateral interest, and FX changes. Assume no earlier liquidation or transfers.

1. Calculate each funding payment or receipt. Then calculate profit after funding and the return on initial collateral.
2. Alternatively, after payment 3, the price used to value the position falls to USD 88. Required maintenance margin is 10% of current position value. Does the account value meet this minimum?

Solution: Funding, Leverage, and Liquidation

[Background Context]

Calculation	USD amount
Long pays after interval 1: $100(100)(0.0001)$	1.00
Long pays after interval 2: $100(102)(0.0002)$	2.04
Long receives after interval 3: $-100(101)(-0.0001)$	1.01
Net funding cost	2.03
Price P&L: $100(104-100)$	400.00
P&L after funding	397.97

Return on initial collateral: $397.97/2{,}000 = \mathbf{19.90}\%$. The price rose 4%, but initial exposure was five times the margin deposit.

At USD 88, account equity is $2{,}000 + 100(88-100) - 2.03 = \mathbf{797.97}$. Required maintenance margin is $10\% \times 100 \times 88 = \mathbf{880}$. The account value is below the required margin. Under this rule, the platform liquidates the position.

Global Equity Access: Ownership and Derivative Exposure

Route	Investor's claim	Costs and contractual questions
Foreign share / ADR	Share or receipt	Custody, access, FX conversion, dividends, and taxes.
ETF	Fund units	Expenses, replication, domicile, and FX exposure.
Equity swap	Swap claim	Financing, dividends, collateral, dealer risk, and closeout.
Equity perp	Perp position	Funding, mark price, liquidation, collateral, and venue rules.

An Exchange-Traded Fund Holds an Investment Portfolio [Core Concept]

An **exchange-traded fund (ETF)** combines investors' money. Its units trade on an exchange. Its **portfolio** is the set of investments it holds or gains exposure to through contracts.

- Buying a fund unit gives an interest in the fund. The **underlying assets** are the shares, bonds, or other investments determining its value.
- An **index** measures the performance of a specified set of securities using stated weights. An index-tracking fund aims to follow it. An actively managed fund lets a manager choose holdings under its mandate.
- A fund's **trading counter** is a currency in which its units can trade.

ETF Investors and Authorized Participants [Core Concept]

- A retail investor normally buys or sells ETF units on an exchange. The fund holds securities or obtains the stated investment exposure.
- Net asset value (NAV) per unit is the fund's asset value minus liabilities, divided by units outstanding. The exchange price is the price another trader will pay or accept now.
- An **authorized participant (AP)** can create or redeem large blocks of ETF units directly with the fund. It exchanges the required securities or cash for those units.
- A basket is the specified package of underlying securities. An AP can trade when the ETF price differs enough from the basket's value. Creation or redemption connects the two prices. Fees and hedging risk limit the profit.

ETF structure and history →

Historical ETF size → Hong Kong ETF history → Mainland ETF history →

ETF Creation: The Trading Opportunity

[Background Context]

An AP can buy the exact basket needed to create 50,000 ETF units for USD 2.5M. It can sell each new unit for USD 50.50. Assume the AP can trade at both stated prices. Ignore costs. Describe the transfers and calculate the AP's profit.

1. Which assets does the AP buy, and what does it deliver to the fund?
2. How many ETF units does it receive, and what are the sale proceeds?
3. Calculate the profit. How do the trades affect the ETF premium?

ETF Creation: A Price above the Asset Value [Core Concept]

Illustration: an ETF trades at \$50.50 per share, but its underlying basket is worth \$50 per share. Creation unit: 50,000 shares. Ignore fees, spreads, taxes, and price changes.

1. Buy the required basket for $50{,}000 \times 50 = \$2{,}500{,}000$.
2. Deliver the basket to the fund and receive 50,000 ETF shares through an authorized participant's creation process.
3. Sell those shares for $50{,}000 \times 50.50 = \$2{,}525{,}000$.
4. Gross difference is \$25,000. Basket purchases and ETF sales tend to narrow the premium. Real costs and risk limit the trade.

Redemption reverses the direction when the ETF trades below basket value. Some funds use cash creation/redemption. ETFs may be active or passive. Exchange trading does not determine the investment strategy.

← Main slide

An ETF Price below the Previous NAV

[Background Context]

It is morning in Hong Kong and the market for the underlying US shares is closed. A HK ETF's last published NAV is HKD 100 per unit. Its current bid is HKD 97. Related futures prices suggest a current portfolio value of HKD 97.50 per unit. A futures contract specifies a trade for a later date.

1. Split the HKD 3 gap into the estimated fall in asset value and the extra cost of selling now.
2. For 40,000 units, calculate sale proceeds and the shortfall relative to estimated current value.
3. Explain why buying at 97 and "redeeming at 100" may be impossible.

Execution checklist →

Valuation	Per unit	40,000 units
Yesterday's NAV	100.00	HKD 4,000,000
Estimated current value	97.50	HKD 3,900,000
Executable bid	97.00	HKD 3,880,000

- HKD 100,000 reflects the estimated market decline. A further HKD 20,000 is the gap between current value and sale proceeds.
- The execution gap is $0.50/97.50 = 0.513\%$ of current value, or 0.50% of yesterday's NAV. Specify the denominator.
- Redemption delivers current basket value under the fund's rules. The redemption value can differ from yesterday's NAV.

Portfolio Weights and the Spending Goal

[Core Concept]

- An **asset allocation** divides a portfolio among categories such as equities, bonds, and cash. A **weight** is a holding's current value divided by the portfolio value.
- A **defensive allocation** aims to limit losses and pay for planned spending. Cash and high-quality bonds can serve this role. Check their currency risk, credit risk, and interest-rate risk.
- An **emergency reserve** is money kept available for unexpected spending. A **time horizon** is the period until the money is needed.
- A **benchmark** is a stated reference portfolio for comparing allocation or performance. A policy sets the investor's targets and rules for changing holdings.

For this example, HKD 60,000 equities divided by the HKD 100,000 long-term portfolio is a 60% weight.

Source: Further reading: Yale Insights, James Choi's research on how much to invest in stocks (2026).

Example: You have started your first full-time job in Hong Kong and earn HKD. You want to invest HKD 100,000 for at least ten years.

- You can add HKD 3,000 each month. You may study overseas in a few years. You have not chosen a country or committed to the expense.
- First, decide how much money to invest for the long term. Keep a separate plan for earlier spending.
- For long-term investments, choose the equity weight, country weights, and currency policy for defensive assets.
- Then compare funds and decide how to place trades.

Example: Job Income and Investment Risk [Core Concept]

The table gives possible one-year outcomes for a Hong Kong real estate sector worker. Invest HKD 100,000 in either home or foreign equities. Foreign returns already include FX conversion to HKD.

State	Probability	Job income (HKD)	Home return	Foreign return
Expansion	60%	300,000	20%	10%
Local property downturn	20%	180,000	−30%	5%
Global recession	20%	240,000	−25%	−30%

1. Calculate each investment's expected return.
2. In the local downturn, add annual job income to ending investment value. Which investment supports spending better?

Example: Equal Expected Returns and Different Household Risks [Core Concept]

Both investments have a 1% expected HKD return under the stated probabilities.

Income + ending investment value	Home equities	Foreign equities
Expansion	HKD 420,000	HKD 410,000
Local property downturn	HKD 250,000	HKD 285,000
Global recession	HKD 315,000	HKD 310,000

- For a worker whose pay depends on foreign technology sales, the foreign equity portfolio could instead add to the same business exposure.
- Investment risk depends on how returns move with income and spending needs.

Example: Equity Allocation across the Full Distribution [Core Concept]

The table shows an example of the possible outcomes over one year. Defensive assets (for example, a deposit account) return 2% in HKD in every state. Equity returns below are also in HKD.

State	Probability	Equity return	40% equity	60% equity	80% equity
Normal	60%	15%	7.2%	9.8%	12.4%
Boom	20%	35%	15.2%	21.8%	28.4%
Recession	20%	−40%	−14.8%	−23.2%	−31.6%
Expected return		8%	4.4%	5.6%	6.8%

$$R_p = wR_e + (1 - w)(2\%).$$

More equity increases both expected wealth and the recession loss. For HKD 100,000, compare expected ending wealth with the cash needed if the recession occurs.

Markets Included in Global Indexes

[Background Context]

Index	Included markets	Size coverage
MSCI World	Developed markets	Large and medium-sized listed companies
MSCI ACWI	Developed and emerging markets	Large and medium-sized listed companies

- A fund tracking MSCI World excludes emerging markets. An investor seeking both needs broader coverage or a separate emerging-market holding.
- Adding an emerging-market fund to an ACWI fund increases the existing emerging-market weight.
- MSCI is the index provider. ACWI means All Country World Index.

Source: MSCI, World and ACWI index descriptions, accessed October 22, 2026.

Global markets and investors →

Home Bias Relative to a Global Benchmark

[Background Context]

Suppose home-market shares are 5% of an investor's chosen global equity benchmark, but 80% of the investor's equity holdings.

$$\text{Home overweight} = 80\% - 5\% = 75 \text{ percentage points.}$$

- Familiar firms, easier access, information costs, taxes, and institutional allocation regulations can influence the investor's choices.
- Home-currency spending can justify cash or bonds matched to that spending. But home equities might fall when local job income falls (positive correlation, vs negative for hedging / diversification)

Diversification mathematics → Overlap and exposure →

Illustration: the investor places the equity allocation in 50% global fund, 30% US fund, and 20% emerging-market fund. Assume the global fund is 60% US, 30% other developed, and 10% emerging.

Exposure	Calculation within the equity allocation	Result
US	$50\% \times 60\% + 30\%$	60%
Other developed	$50\% \times 30\%$	15%
Emerging markets	$50\% \times 10\% + 20\%$	25%

- Emerging markets are 25% of HKD 60,000 equities: HKD 15,000, or 15% of all savings. The benchmark amount is HKD 6,000.
- For any exposure c, calculate $w_c = \sum_j w_j a_{jc}$, where w_j is fund j's portfolio weight and a_{jc} is its share in region c.

From Week 4 FX Risk to Portfolio Return Risk

[Background Context]

We now apply the Week 4 return-risk measures to investment holdings. Use one currency and one return horizon throughout.

Symbol / term	Meaning
$E[R]$, expected return	Probability-weighted average of possible returns.
σ, volatility	Standard deviation: the scale of returns around their mean.
σ^2, variance	Volatility squared.
ρ, correlation	How closely returns move together, on a scale from -1 to $+1$. Covariance is $\rho\sigma_1\sigma_2$.

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2.$$

Connection: The w terms are value weights. When returns move differently, combining investments can reduce volatility. The portfolio can still lose money.

Diversification under Different Correlations [Core Concept]

Two equity markets each have 20% volatility, measured in HKD over the same horizon. Hold 50% in each.

$$\sigma_p = \sqrt{(0.5)^2(0.2)^2 + (0.5)^2(0.2)^2 + 2(0.5)(0.5)\rho(0.2)(0.2)}.$$

Assumed correlation ρ	0.30	0.80	1.00
Portfolio volatility	16.12%	18.97%	20.00%

- Lower correlation reduces portfolio volatility in this example. It does not prevent both markets from losing money together.
- Repeat the calculation with higher correlations and higher individual volatilities. Historical estimates can change.

Variance Arithmetic →

Historical bond returns → Historical correlations → Estimation uncertainty →

Diversification When Both Markets Become Riskier [Core Concept]

The upcoming game has two states of the world. Returns are in HKD.

State	Probability	Home mean	Global mean	Volatility / correlation
Normal	85%	12%	11%	20% each / 0.30
Stress	15%	−30%	−20%	30% each / 0.90

$$\sigma_{50/50} = \sigma\sqrt{(1+\rho)/2}: \quad 20\%\sqrt{0.65} = 16.12\%, \quad 30\%\sqrt{0.95} = 29.24\%.$$

- Volatility rises by about 1.81 times. Individual markets become riskier and move more closely together.
- A 50/50 mix minimizes variance within each stated year type here. Tail losses are outcomes near the worst end of the distribution. Lower home weight can reduce those losses because home returns are worse in stress.

Expected Return across Normal and Stress Years

[Background Context]

For an equal mix of home and global equities, before fund costs:

$$E[R \mid \text{normal}] = (12\% + 11\%)/2 = 11.5\%, \qquad E[R \mid \text{stress}] = (-30\% - 20\%)/2 = -25\%.$$

$$E[R] = 0.85(11.5\%) + 0.15(-25\%) = 6.025\%.$$

- The vertical bar is "conditional on the given state." The full expected return weights both year types by their probabilities.
- The game's **mixture model** combines normal-year and stress-year return distributions. A **single-normal model** uses one symmetric, bell-shaped return distribution fitted to the overall mean and volatility.
- Two models can have the same mean and volatility but different probabilities of large losses. Compare their worst outcomes as well as their averages.

Written check: The stress probability rises to 20%. The mean return within each year type stays unchanged. Calculate the new overall mean. Explain the change.

The Capital Asset Pricing Model (CAPM) [Core Concept]

Investors can spread firm-specific risk across many companies. CAPM prices an asset's exposure to the risk of the market portfolio.

$$E[R_i] = r_f + \beta_i\big(E[R_M] - r_f\big).$$

- r_f: the risk-free return over the same horizon and in the same currency as the asset return.
- $E[R_M] - r_f$: the market's expected excess return, or **market risk premium**.
- β_i: how strongly the asset's return moves with the market return. Higher beta requires a higher expected return when the market premium is positive.

Week 5 discounted uncertain cash flows at a required return. CAPM provides one model for that required return.

Beta: Exposure to Market Risk

[Core Concept]

Estimate beta using asset and market returns measured in the same currency and over the same periods.

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_M)}{\mathrm{Var}(R_M)}.$$

- **Covariance** measures how the asset and market returns move together. Dividing by market variance expresses the sensitivity per unit of market movement.
- Suppose covariance is 0.024 and market variance is 0.020. Then beta is $0.024/0.020 = 1.2$.
- In the fitted model, market excess return rises by 1 percentage point. The asset's predicted excess return rises by 1.2 percentage points. Other news also affects the asset.

Regression details →

Which Market Does the CAPM Use?

[Core Concept]

The asset's beta and the market premium must refer to the same market portfolio.

Model assumption	Meaning and benchmark
Local / segmented market	Barriers limit international investing. Use the relevant local market.
Global / integrated market	Investors can share risk internationally. Use a world market portfolio.
Multifactor extension	Add defined sources of shared risk, such as currency risk. Each has an exposure and a risk premium.

- Market integration describes access and pricing across borders.
- Use the same return currency for the asset, market, and risk-free rate. A country's risky government-bond yield includes credit risk.

CAPM reference → Model comparison → Multifactor model →

CAPM: A Required Return in HKD

[Background Context]

A Hong Kong analyst values a foreign equity investment in HKD. Assume integrated markets and the following inputs:

Input	Value
HKD risk-free rate	3%
World-market premium, using HKD returns	5%
Beta, estimated using HKD returns	1.2

$$E[R_i] = 3\% + 1.2(5\%) = \mathbf{9\%}\ \textbf{in HKD}.$$

This is an expected return required by the model. An actual year's return can be higher or lower. Which inputs would you revisit if you valued the same investment in JPY?

How an ETF Tracks Its Index

- **Full physical replication**: the fund buys the securities in the index at the required weights.
- **Sampling**: the fund holds a subset chosen to reproduce the index's returns. Differences in holdings can cause tracking differences.
- **Synthetic replication**: the fund uses a swap to receive the index return. Performance depends on the swap contract, collateral, and the counterparty meeting its obligations.
- A **prospectus** sets out the legal terms. A **factsheet** summarizes the strategy, holdings, costs, and results.

An interest-rate swap exchanges interest payments. An ETF total-return swap exchanges an investment return.

Fund Domicile, Trading Currency, and Share Class

A fund can have its legal home in Ireland. Its units can trade in USD in London while it holds Japanese shares.

Attribute	Meaning
Domicile	The jurisdiction governing the fund's legal structure.
Listing venue	The exchange where investors trade the fund's units.
Trading currency	The money used to pay for units on a particular counter.
Share class	A version of the fund with stated fees, distribution policy, and any currency hedge.

An accumulating class retains income. A distributing class pays it out. Compare both using total returns and the same reinvestment assumptions.

Reading an ETF Factsheet

[Background Context]

Field	Decision it informs
Index and holdings	Which markets, sectors, and companies will the fund add?
Currency and hedging	Which currency risks remain? What currency does the hedge target?
Replication	Does the fund hold securities, sample them, or use swaps? What counterparty exposure remains?
Ongoing charges and tracking	How much has NAV total return differed from the stated benchmark?
Trading and fund size	What are the bid–ask spread, order size, and risks of fund closure?
Domicile and distributions	What is the fund's legal home? Does it retain income or pay it out?

Read the product key facts statement and prospectus for the exact share class. Check broker access and charges before comparing candidates.

Source: IFEC, ETF key features.

Fund Choice Depends on the Holding Period

[Background Context]

Two funds track the same index with the same tax and currency treatment. Charges remain constant. Ignore compounding and changes in investment value.

Cost as a fraction of investment	Fund A	Fund B
Annual ongoing charge	0.10%	0.25%
One purchase plus one eventual sale, including FX	0.60%	0.10%

1. Find the total cost over one year and over five years.
2. Find the holding period at which the two approximate costs are equal.

Small contributions →

Fund Costs over One and Five Years

Let T be the holding period in years. Total cost is a fraction of the initial investment.

$$C_A(T) = 0.60\% + T(0.10\%), \qquad C_B(T) = 0.10\% + T(0.25\%).$$

Holding period	Fund A	Fund B	Lower approximate cost
1 year	0.70%	0.35%	B
5 years	1.10%	1.35%	A

$$T^* = \frac{0.60 - 0.10}{0.25 - 0.10} = \mathbf{3.33\ years}.$$

- Fund A's combined purchase and sale cost is 0.50 percentage points higher. Its annual charge is 0.15 percentage points lower.
- Different order sizes, taxes, tracking, or currency policies would require a new comparison.

Illustration: a benchmark earns 8.0% over a year and an ETF's NAV total return is 7.7%, in the same currency and with comparable dividend treatment.

- Define **tracking difference** as fund return minus benchmark return. Here it is $7.7\% - 8.0\% = -0.3$ percentage points.
- **Tracking error** measures variability in the fund's return deviations over time. A fund can earn a consistently lower return and still have low tracking error.
- The published NAV return already deducts ongoing fund expenses. Use it directly when calculating the investor's return.
- The investor also pays broker charges and currency-conversion costs. Trading prices can differ from NAV. These costs and price differences affect the investor's return.

Source: IFEC, Tracking difference and tracking error of ETFs.

Dividend Withholding and Fund Costs

[Background Context]

A fund wrapper is the legal vehicle used to hold an investment. Dividend withholding is tax deducted before dividends reach the fund. The game assumes a 2% dividend yield and the withholding rates below.

Game wrapper	Annual fee	Withholding	Annual cost
US-listed ETF	0.05%	30%	$0.05\% + 0.30(2\%) = 0.65\%$
Ireland-domiciled ETF	0.10%	15%	$0.10\% + 0.15(2\%) = 0.40\%$
HK-listed feeder fund	0.25%	30%	$0.25\% + 0.30(2\%) = 0.85\%$

- On HKD 1M, gross dividends are HKD 20,000. The assumed 30% withholding deducts HKD 6,000.
- A feeder fund invests through another fund. The table uses the game's tax and fee assumptions.

Distributions and domicile →

Example: A Foreign Asset Return in HKD

[Background Context]

Invest HKD 850 in an unhedged euro asset. Spot today is 8.50 HKD/EUR, so the investment buys EUR 100. The asset earns an 8% EUR total return. One year later spot is 7.65 HKD/EUR.

1. Calculate the final EUR amount and convert it to HKD.
2. Calculate the HKD return using the initial HKD 850 as the denominator.
3. Predict whether using the fund's HKD trading counter would change this exposure.

USD investor example →

$$\text{EUR payoff} = 100(1.08) = 108, \qquad \text{HKD payoff} = 108(7.65) = 826.20.$$

$$R_{\text{HKD}} = \frac{826.20}{850} - 1 = (1.08)(0.90) - 1 = -\mathbf{2.8}\%.$$

- The general relation is $1 + R_H = (1 + R_F)(S_1/S_0)$, with S in home currency per foreign currency unit.
- Adding 8% and -10% misses the -0.8% interaction term.
- Paying for fund units in HKD leaves the underlying EUR exposure unchanged. A currency hedge is a separate contract.

Source: Trading-counter distinction: IFEC, Multiple Counter Model for ETFs.

Currency Policy for Different Portfolio Roles [Core Concept]

- **Known spending**: hold GBP for a known GBP tuition bill, with funds available on its due date.
- **Defensive bonds**: compare HKD bonds with foreign bonds hedged into HKD. Currency movements can be larger than the bond income.
- **Long-term equities**: compare hedged and unhedged exposure using costs, covariance, and the investor's liabilities.
- The **hedge ratio** is the share of currency exposure covered. A USD-hedged share class targets USD, leaving the HKD investor to assess USD/HKD exposure.

Forward hedge example →

Source: Vanguard, Discussion of currency risk in international stocks and bonds.

A Forward Hedge on the Initial Foreign Investment [Core Concept]

Invest EUR 100 for HKD 850. The asset later pays EUR 105. At the start, sell EUR 100 forward at 8.40 HKD/EUR. At maturity, spot is 7.65 HKD/EUR.

Component	HKD value at maturity
Asset without hedge	$105(7.65) = 803.25$
Short-forward payoff	$100(8.40 - 7.65) = 75.00$
Combined value	$803.25 + 75.00 = 878.25$

$$R_H = 878.25/850 - 1 = \mathbf{3.32}\%.$$

The forward covers the initial EUR 100. Convert the EUR 5 gain at the final spot rate. If the EUR 105 payment were known at the start, selling all EUR 105 forward would fix the receipt at HKD 882.

A Hedge Can Require Cash When the Asset Gains [Core Concept]

Suppose you hold Japanese bonds and sells JPY forward. The hedge ratio is the fraction of the foreign exposure covered. Let H be the covered amount, measured at its initial HKD value. Let g_{JPY} be the percentage rise in JPY against HKD. The assumed one-year forward premium is 3.5%.

$$\text{HKD settlement payment} = H(g_{\mathrm{JPY}} - 3.5\%).$$

- If $H = \mathrm{HKD}\ 20\mathrm{M}$ and JPY appreciates 10%, the foundation pays $20(0.10 - 0.035) = \mathrm{HKD}\ 1.3\mathrm{M}$ on the forward.
- Its JPY asset gains HKD value from appreciation. That value may still be invested when the cash settlement falls due.
- With only HKD 1M cash, it must find HKD 0.3M through a sale or other financing. The hedge reduces currency exposure but changes cash needs.

The game uses net HKD settlement and a simple return approximation. Actual contracts specify delivery, collateral, and payment dates.

Hedged Japanese Bond Return in the Game

Use the game's assumed short rates and quoted basis convention:

$$\frac{F - S_0}{S_0} \approx i_{\mathrm{HKD}} - i_{\mathrm{JPY}} + b = 3.5\% - 0.5\% + 0.5\% = 3.5\%.$$

$$\text{Income plus forward adjustment} = 1.0\% + 3.5\% = \mathbf{4.5}\%.$$

- Here S and F are HKD per JPY. The game defines positive b as increasing the premium received when selling JPY forward.
- This approximation ignores the interaction term, bond-price movements, credit, and implementation costs. The short-rate gap and the basis set the forward premium.
- In the game's stress years, JPY rises and increases the unhedged asset's HKD value. A hedge reduces this gain and can require cash payments.

Modified duration summarizes how much a bond's price changes when its yield changes. For a small yield change:

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y.$$

- P is the initial price. $\Delta P/P$ is its proportional change. Δy is the yield change in decimal units: one percentage point is 0.01.
- With modified duration 7, a one-percentage-point yield rise implies about a 7% price fall, holding other inputs fixed.
- Coupon income and currency conversion are separate parts of the investor's total return. We combine them after estimating the price effect.

Illustration: a foreign bond portfolio starts with a 5% yield and modified duration 7. Yields rise by 1 percentage point over the year.

$$\frac{\Delta P}{P} \approx -D_{\text{mod}}\Delta y = -7(0.01) = -7\%.$$

- Approximate local return: 5% assumed income -7% price change $= -2\%$. Assume no default. This linear estimate ignores changes in price sensitivity and the effect of time passing.
- If the currency also falls 10%, the unhedged HKD return is approximately $(0.98)(0.90) - 1 = -11.8\%$.
- Credit losses and a difficult sale can add to the loss. Bonds exposed to the same bad states as equities provide less protection in those states.

Portfolio role: Assess duration, credit quality, and currency exposure together when choosing defensive assets.

Rebalancing after an Equity Gain

[Core Concept]

Rebalancing restores the target portfolio weights through trades, contributions, or withdrawals. The long-term portfolio starts with HKD 60,000 equities and HKD 40,000 defensive assets. Equities gain 20%, defensive assets return 0%, and there are no contributions or withdrawals.

1. Find both ending values and the new equity weight.
2. Calculate the trade needed to restore exactly 60/40 with no new savings.
3. Instead of selling, find the new savings needed if all new money goes to defensive assets.

Performance benchmark →

Withdrawal practice →

Rebalancing after Relative Returns Change [Core Concept]

Illustration: begin with HKD 60,000 equities and HKD 40,000 defensive assets. Equities gain 20%, defensive assets return 0%, and there are no contributions or withdrawals.

Holding	Starting value	Ending value	Target at 60/40
Equities	60,000	72,000	67,200
Defensive assets	40,000	40,000	44,800
Total	100,000	112,000	112,000

- New equity weight: $72{,}000/112{,}000 = 64.29\%$. Sell HKD 4,800 of equities and buy defensive assets to restore 60/40.
- Alternatively, add HKD 8,000 of new savings to defensive assets. Then equity weight is $72{,}000/120{,}000 = 60\%$.

A Withdrawal after a Market Loss

[Background Context]

A portfolio begins at HKD 60,000 equities and HKD 40,000 defensive assets. Equities fall 25% and defensive assets return 0%. The investor then withdraws HKD 10,000 from defensive assets.

1. Calculate portfolio weights immediately before and after the withdrawal.
2. Find the trade needed to restore 60/40 after the withdrawal. Explain why the trade differs from rebalancing before the withdrawal.

Answer →

Withdrawal and Rebalancing: Answer

Position	Equities	Defensive assets	Equity weight
After the market loss	HKD 45,000	HKD 40,000	$45/85 = 52.94\%$
After the withdrawal	HKD 45,000	HKD 30,000	$45/75 = 60\%$

- The withdrawal restores the exact target in this example. No rebalance trade is needed afterwards.
- Before the withdrawal, restoring 60% equities would require $0.60(85{,}000) - 45{,}000 = 6{,}000$ of equity purchases.
- Planned cash flows can change the trade required. Evaluate the portfolio after known withdrawals and contributions.

A Rebalancing Rule Chosen in Advance

[Background Context]

- **Example policy**: review once a year and after a major change in goals. Rebalance if the equity weight lies outside a 55–65% band around a 60% target.
- A 64.29% equity weight stays inside this band, so this rule requires no trade. A different rule could restore exactly 60%.
- First, add new savings and cash distributions to holdings below their target weights. Compare remaining trades with transaction costs and relevant tax effects.
- Rebalancing keeps portfolio weights close to their targets. Assets sold can continue to rise, so rebalancing can reduce returns.

Source: Investor.gov, Asset Allocation and Diversification.

A Joint Equity and Currency Shock

[Core Concept]

Assume 60% foreign equities and 40% defensive assets. Foreign equities fall 30% in their local currency. That currency falls 10% against HKD. Defensive assets lose 2% in HKD.

1. Find the foreign equity return in HKD.
2. Find the return and ending value of a HKD 100,000 portfolio.
3. The investor also loses three months of income. Explain which assets can fund spending without selling equities after the fall.

A Joint Equity and Currency Stress Test

[Core Concept]

Shock: 60% in foreign equities and 40% in defensive assets. Foreign equities fall 30% in local currency. Their currency falls 10% against HKD. Defensive assets lose 2% in HKD.

$$R_{\text{equity,HKD}} = (0.70)(0.90) - 1 = -37\%,$$

$$R_p = 0.60(-37\%) + 0.40(-2\%) = -\mathbf{23}\%.$$

- HKD 100,000 becomes HKD 77,000 in this stated scenario. Other outcomes may be better or worse.
- Could the investor still cover planned spending if job income also fell? Would the cash reserve allow the investor to avoid selling after the decline?
- A cash shortfall calls for a larger reserve, less equity risk, or a revised spending plan.

Class Exercise: A Portfolio with a GBP Tuition Goal

New information: the investor has HKD 100,000 after setting aside an emergency reserve. The investor must now pay GBP 4,000 tuition in one year. Spot is 10 HKD per GBP.

- For this exercise, set aside the full tuition amount in GBP cash today. Ignore deposit interest, conversion charges, and inflation.
- Invest the remaining amount for ten years using a 60/40 equity/defensive policy. Use the assumed 60/30/10 regional split within equities.
- Calculate each allocation in HKD. Explain the currency policy for tuition and for the long-term defensive assets.
- Propose a rebalancing rule. Identify one overlap between funds or one trading-cost problem to examine before investing.

Work individually. Then compare your answer with the solution. State all denominators: total savings, long-term portfolio, or equity allocation.

Answer →

Tuition and Long-Term Allocation

[Background Context]

Tuition: buy GBP 4,000 for HKD 40,000. The remaining HKD 60,000 funds the long-term portfolio.

Holding	HKD value	% of long-term fund	% of all savings
GBP tuition cash	40,000	—	40.0%
US equities	21,600	36%	21.6%
Other developed equities	10,800	18%	10.8%
Emerging-market equities	3,600	6%	3.6%
Long-term defensive assets	24,000	40%	24.0%

- Total equities are HKD 36,000: 60% of long-term assets, or 36% of all HKD 100,000. Tuition has its own currency and horizon.
- For assumed future HKD spending, assess duration, credit, and FX risk in the HKD defensive allocation. Apply rebalancing only within the long-term portfolio.

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The Foundation's Portfolio and Payment Obligations

[Background Context]

In this game, you manage HKD 100M for a Hong Kong foundation. The foundation invests to support future activities. It needs investment returns for spending and cash for currency-hedge payments.

- Choose the equity share, home share within equities, foreign-equity wrapper, JPY hedge ratio, and HKD cash reserve.
- After equities and cash, split the remainder equally between HKD bonds and Japanese government bonds.
- Example: 60% equities and HKD 5M cash leave HKD 35M for bonds, or HKD 17.5M in each bond allocation.
- If 10% of the equity allocation is home equity, home equity is HKD 6M. The denominator is the HKD 60M equity allocation.

Mandate: Choose the highest expected return among portfolios that pass both risk limits.

Expected Shortfall (ES): Average of the Worst Returns [Core Concept]

Week 4 defined value at risk (VaR) as a loss cutoff. Here use signed returns: more negative means a larger loss. Suppose 100 equally likely one-year scenarios have already been ranked. The five worst returns are:

Rank from worst	1	2	3	4	5
Return	-40%	-35%	-30%	-28%	-22%

$$ES_{5\%} = \frac{-40 - 35 - 30 - 28 - 22}{5}\% = -\mathbf{31}\%.$$

- Using the fifth-worst observation as the empirical cutoff, 5% VaR is -22%. ES averages outcomes in that lower tail.
- The game reports signed returns. ES of -30% passes an "at least -31%" limit. ES of -35% fails.

Check: If only the worst return becomes -50%, which statistic changes under this convention?

Tail Loss and Cash Shortfall

[Core Concept]

- **Tail-loss limit**: the average return in the worst 5% of outcomes must be at least -31% in the game.
- **Cash limit**: the probability that hedge settlement exceeds the cash reserve must be at most 2%.
- Suppose hedge payments exceed the cash reserve in three of 100 equally likely scenarios. The 3% shortfall probability exceeds the 2% limit.
- A hedge can reduce one exposure and still worsen another outcome. In the game, it removes JPY appreciation gains in stress and may require cash settlement.

Model check: A single normal distribution can have the same mean and volatility as the mixture model. Their probabilities of large losses can differ. Compare both models before choosing a policy.

A Paper Portfolio Committee: Apply Both Limits

The committee must meet ES $\geq -31\%$ and cash-shortfall probability $\leq 2\%$. These outputs refer to otherwise feasible policies.

Policy	Expected return	ES	Cash shortfall probability
A	5.5%	−34%	1.0%
B	5.0%	−30%	2.8%
C	4.8%	−30.5%	1.5%
D	4.5%	−29%	0.8%

Decision memo: Choose a policy that meets the stated objective and both limits. Explain why you reject each alternative. Explain how more cash could change expected return and shortfall risk.

Answer →

Applying Both Limits: Answer

[Background Context]

- Policy C has the highest expected return among policies that meet both limits.
- A fails the ES limit despite meeting the cash limit. B meets the ES limit but fails the cash limit.
- D passes both, but its expected return is lower than C's under the stated objective. A different mandate could justify choosing D.
- Adding cash may lower expected return if it replaces a higher-return asset. It can reduce forced sales by covering settlement. Recalculate expected return and both risk measures.

Further check: If the cash limit tightens to 1%, which listed policy remains feasible? Answer: D.

In-Class Game 9: Portfolio Risk Committee

[Background Context]

Role: manage a Hong Kong foundation's HKD 100M for ten years. It pays HKD 4M of grants each year. Work individually at your own pace.

- **Meetings**: in Years 1, 4, and 7, choose equity and home shares, fund type, JPY hedge ratio, and cash reserve.
- **Analyst requests**: calculate stress-year volatility, fund cost, hedged yield, and hedge settlement. Your answers give access to the dashboard.
- **Urgent day**: before Year 3, raise HKD 4M while the US market is closed.
- **Score**: final fund value plus grants paid, less penalties and analyst fees. Each grant cut costs 5 points. Each meeting that breaches a board limit costs 2 points. Limits: ES $\geq -31\%$ and cash-shortfall probability $\leq 2\%$. One point equals HKD 1M.

Open Week 9 Game

Game solution →

Committee Debrief: Why a Higher Mean Can Fail

Policy	Passing choice	Single-normal choice
Equity allocation	60%	70%
Home share within equities	10%	10%
JPY hedge ratio / cash reserve	100% / HKD 5M	100% / HKD 3M
Mean return	5.17%	5.37%
Full-model ES	−30.8%	−36.7%

Both use the game's Ireland-domiciled wrapper. The passing choice has a 0.3% cash-shortfall probability. The portfolio with the higher expected return fails the mixture model's ES limit.

Explain your decision: Which alternative did you reject, and what evidence caused you to reject it?

Full game answer → Paper committee case →

Consider a Japan-based pension fund that spends JPY and owns HKD bonds without a currency hedge. Assume the bond earns 3.5% in HKD and JPY appreciates 10% against HKD.

$$\frac{S_1}{S_0} = 1.10 \quad \text{for } S \text{ in HKD/JPY}, \qquad R_{\text{JPY}} = \frac{1.035}{1.10} - 1 = -\mathbf{5.91}\%.$$

- Each HKD now converts into fewer JPY. JPY appreciation helps an HKD investor with JPY assets. It hurts a JPY investor with HKD assets.
- Assess defensive assets using the investor's spending currency, payment dates, and other risks.

Exit response: Explain one choice you changed during the game and write one policy rule that addresses the risk you found.

Example investment policy →

Final Written Check: Claims, Hedging, and Returns

1. A firm receives USD 100M from a new bond issue and enters a same-currency interest-rate swap. Which contract supplies the funding? Which changes net interest payments?

2. An ETF trades in HKD and holds unhedged foreign shares. The shares earn 8% in their own currency and that currency falls 10% against HKD. Calculate the HKD return.

3. A portfolio has HKD 60,000 equities and HKD 40,000 defensive assets. It must first reserve HKD 40,000 for tuition. If the remaining portfolio keeps a 60% equity target, how many HKD belong in equities?

Individual work: Write each result and a short explanation. Keep earlier slides closed. Ignore costs, taxes, and tracking differences.

1. The **new bond issue** raises funds from bondholders. The **swap** exchanges interest cash flows and changes net rate exposure. The issuer still owes the bond's promised payments.

2. The total HKD return is $(1.08)(0.90) - 1 = \mathbf{-2.8\%}$. The trading counter states how the ETF trade settles. It adds no currency hedge by itself.

3. Tuition leaves $100{,}000 - 40{,}000 = 60{,}000$ for long-term investment. The equity target is $0.60(60{,}000) = \mathbf{HKD\ 36{,}000}$. The target's denominator is the remaining portfolio.

Self-check: Mark any answer that used the wrong contract, exchange-rate operation, or portfolio denominator. Write the corrected rule beside it.

Summary and Course Progress

- International bonds and equities expose investors to different combinations of market, credit, liquidity, and currency risk.
- Portfolio construction begins with goals and spending currencies. Equity allocation determines how much market risk the investor takes.
- Global benchmarks clarify country weights. Combining funds requires checking overlapping holdings and the resulting economic exposure.
- Fund selection requires a stated currency policy and a comparison of total costs. A rebalancing rule keeps decisions tied to the investment policy.

Next week: payment systems and the dollar's international role, including settlement risk and the liquidity needed to move money across borders.

Macaulay and Modified Duration

[Background Context]

For a fixed-rate bond with annual coupons and no embedded early-repayment option, **Macaulay duration** is the present-value-weighted average payment date:

$$D = \frac{\sum_{t=1}^{T} \left[\frac{tC}{(1+y)^t} \right] + \frac{TF}{(1+y)^T}}{P}, \qquad D_{\text{mod}} = \frac{D}{1+y}.$$

Here C is the annual coupon, F face value, T final year, y annual yield, and P price. Each payment's present value divided by P is its weight.

- Modified duration gives the local price sensitivity $\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y$. Longer payment timing usually increases that sensitivity.
- A floating coupon updates at a reset date, reducing exposure to changes in the reference rate.
- A cap is a maximum rate and a floor a minimum rate. These limits can change how a floating payment responds to rates.

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How Government Credit Risk Can Reach Companies

[Background Context]

A **sovereign** borrower is a national government. Its credit spread is the yield difference from a comparable benchmark. A downgrade is a rating agency's reduction in its credit assessment.

- **Payment restrictions**: currency-conversion or transfer restrictions can prevent a company from paying foreign creditors even when its business earns cash.
- **Bank losses**: banks holding government bonds can lose value when their yields rise. Weaker bank balance sheets can restrict lending to local firms (Week 7).
- **Government financing pressure**: higher expected taxes or competition for funding can raise company financing costs.

Application: During Europe's sovereign-debt crisis, these channels linked government stress, bank funding, and business credit.

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Company Funding: Calculation Details

[Background Context]

All amounts are USD millions. Each contract raises 100M today and repays principal in Year 5. The common discount rate is 5.2%.

Assumed path	Fixed bond PV	Floating bond PV	Loan PV
3, 4.5, 3.5, 5, 4%	99.14	101.18	100.66
3.5, 3, 2.5, 2, 1.5%	99.14	95.05	96.99

$$PV = \sum_{t=1}^{5} \frac{C_t}{1.052^t} + \frac{100}{1.052^5}.$$

The lender chooses the higher of the previous loan rate and the new benchmark plus 1 point. Falling benchmarks lower the floating bond coupon but may leave the loan rate unchanged.

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Appendix: ADR Arbitrage Step-by-Step

[Background Context]

Assume one underlying share per ADR ($n = 1$), immediate conversion between the share and receipt, and initially ignore fees/taxes. A UK share trades in London at $P_{\text{FS}} = 10.00$ GBP. Exchange rate $S = 1.30$ USD per GBP. Implied ADR price $= 13.00$ USD. Suppose the ADR trades in New York at $P_{\text{ADR}} = 13.50$ USD.

Arbitrage Execution Steps

1. Borrow 13.00 USD and convert to GBP at $S = 1.30 \implies 10.00$ GBP.
2. Purchase 1 underlying share in London for 10.00 GBP.
3. Transfer share to depositary bank to issue 1 ADR.
4. Sell ADR in New York for 13.50 USD.
5. Repay the same-day USD funding. Gross spread is $13.50 - 13.00 = 0.50$ USD per ADR before FX bid–ask, stock/ADR trading, creation, custody, tax, borrow, and timing costs.

With conversion, $P_{\text{ADR}} \approx nP_{\text{FS}}S$, subject to trading and financing costs.

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Two-Asset Portfolio Variance: Let domestic asset weight be w and foreign weight $1-w$. Variances are σ_d^2, σ_f^2 and covariance is $\sigma_{df} = \rho_{df}\sigma_d\sigma_f$.

$$\sigma_p^2 = w^2\sigma_d^2 + (1-w)^2\sigma_f^2 + 2w(1-w)\sigma_{df}$$

The weight giving the lowest return variance (allowing unconstrained weights) is ($w_f^* = 1 - w^*$):

$$1 - w^* = \frac{\sigma_d^2 - \sigma_{df}}{\sigma_d^2 + \sigma_f^2 - 2\sigma_{df}}$$

Numerical Example

- Domestic asset volatility $\sigma_d = 20\%$. Foreign asset volatility $\sigma_f = 20\%$.
- Correlation $\rho_{df} = 0.30 \implies \sigma_{df} = 0.30 \times 0.20 \times 0.20 = 0.012$.
- Minimum variance allocation: $1 - w^* = 50\%$ foreign assets.
- Portfolio volatility at $w = 0.50$: $\sigma_p = 16.12\%$ (vs. 20% in the all-domestic asset).

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Valuing Brazilian Equity: Keep Currency and Model Assumptions Consistent

- USD risk-free rate $R_{f,\text{USD}} = 4.0\%$. Global equity premium $= 5.5\%$. Global beta $\beta_g = 1.20$.
- A Brazilian government bond in USD yields 7.5%. It carries default risk. Its spread includes expected losses, selling costs, and compensation for risk.
- A local Brazilian-real (BRL) model instead uses a defensible BRL default-free proxy, local beta, and local equity premium, with BRL cash flows.

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Global CAPM, in USD: with risk-free rate 4%, global-market premium 5.5%, and global beta 1.2,

$$E[R_i^{\mathrm{USD}}] = 4\% + 1.2(5.5\%) = 10.6\%.$$

A **factor** is a specified common source of return risk. A multifactor extension can add currency and country risk:

$$E[R_i^{\mathrm{USD}}] - r_{f,\mathrm{USD}} = \beta_g \lambda_g + \beta_{\mathrm{FX}} \lambda_{\mathrm{FX}} + \beta_c \lambda_c.$$

- Each β is the asset's exposure to that factor. Each λ is the expected compensation per unit of exposure. Subscripts g, FX, and c refer to global market, currency, and country factors.
- The exposures and premia must be estimated consistently. A government's credit spread also includes expected default losses and selling costs, so it cannot automatically replace λ_c.
- Express cash flows and required returns in the same currency. BRL denotes Brazilian reais, while this example uses USD.

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A Real Bond Issue: Toyota Motor Corporation, June 2025

[Background Context]

The prospectus specifies three USD 500M fixed-rate senior-note tranches:

Maturity	Annual coupon	Full half-year coupon per USD 1,000
2027	4.186%	$1{,}000(0.04186)/2 = 20.93$
2030	4.450%	$1{,}000(0.04450)/2 = 22.25$
2035	5.053%	$1{,}000(0.05053)/2 = 25.265$

Source: Toyota Motor Corporation, 2025 prospectus.

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Next →

Financing Comparison: Payment Rules

Borrow USD 100M for five years. All three contracts repay 100M principal in Year 5. Rates below are annual rates paid once a year.

1. Fixed bond: pay USD 5M every year.
2. Floating bond: pay $100(L_t + 1.5\%)$M. For the main path, coupons are 4.5, 6, 5, 6.5, and 5.5M.
3. Loan: start at $L_1 + 1\%$. Thereafter the lender chooses $r_t = \max(r_{t-1}, L_t + 1\%)$. Coupons are 4, 5.5, 5.5, 6, and 6M.
4. In Year 3 the benchmark falls to 3.5%, but the lender retains 5.5%. This option explains why a lower initial spread can be costly later.

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Financing Comparison: Present Values

Amounts are USD millions. The final cash flow includes principal.

Year	Discount factor	Fixed CF	Fixed PV	Float CF	Float PV	Loan PV
1	0.95057	5	4.753	4.5	4.278	3.802
2	0.90358	5	4.518	6	5.422	4.970
3	0.85892	5	4.295	5	4.295	4.724
4	0.81646	5	4.082	6.5	5.307	4.899
5	0.77611	105	81.491	105.5	81.879	82.267
Total			99.14		101.18	100.66

The loan's final payment is USD 106M. Totals use unrounded values. These are borrower costs conditional on one rate path. The common discount rate is an assumption for this comparison.

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Financing Comparison, Step 3: Costs across Business States

Assume annual operating cash flow and the floating benchmark move together. Debt principal is USD 100M. Ignore principal repayment in this annual liquidity comparison.

State	Cash flow	Benchmark	Fixed coupon	Floating coupon
Strong	140	5.5%	5	7
Normal	120	4.5%	5	6
Weak	100	2.0%	5	3.5

1. Subtract coupon from operating cash flow. Fixed debt leaves 135, 115, and 95. Floating debt leaves 133, 114, and 96.5.
2. Floating debt preserves 1.5M more cash in the weak state, but costs more in strong and normal states.
3. The benefit depends on covariance: if interest rates rise when the firm's cash flow falls, floating debt makes the liquidity problem worse.

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Equity Claims, Market Makers, and Cross-Border Access

[Background Context]

- Common equity has no contractual maturity and receives residual profits through dividends or repurchases. Shareholders can lose their entire investment.
- An initial public offering (IPO) first sells shares to the public. A follow-on offering sells new shares after listing. New shares can dilute existing holders' percentage ownership.
- Dealer and auction features can coexist. Nasdaq developed from a dealer market. NYSE combines electronic order matching with designated market makers. Both markets combine different trading methods.
- Shanghai–Hong Kong Stock Connect began in 2014. Shenzhen–Hong Kong followed in 2016. Exchange trading links and clearing arrangements connect investors to eligible shares. Access remains subject to market rules and quotas.

Stock Connect links do not imply identical trading, ownership, settlement, or capital-control rules across the markets.

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Foreign Equity Return: Convert Back to the Investor Currency

[Background Context]

A USD investor puts \$1,000 into a euro asset. Initial spot is 1.10 USD per EUR. The asset earns 8% in EUR. Final spot is 1.00 USD per EUR. Ignore dividends outside the quoted total return, costs, and taxes.

1. Initial investment: $1{,}000/1.10 = \text{EUR } 909.09$.
2. Final EUR value: $909.09 \times 1.08 = \text{EUR } 981.82$.
3. Final USD value: $981.82 \times 1.00 = \$981.82$.
4. USD return: $981.82/1{,}000 - 1 = -1.818\%$.

$$1 + R_{USD} = (1 + R_{EUR})\frac{S_1}{S_0} = 1.08\frac{1.00}{1.10}.$$

A positive local return can become a loss after currency translation. Adding 8% and the FX percentage change omits the interaction term.

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Diversification: Expand the Two-Asset Variance Formula

[Background Context]

Two assets each have 20% annual volatility. Put half the portfolio in each. Correlation is 0.30. Returns must use the same currency and horizon.

1. Covariance is $\rho\sigma_1\sigma_2 = 0.30(0.20)(0.20) = 0.012$.
2. Own-variance terms are $(0.5)^2(0.2)^2 = 0.01$ each.
3. The covariance term is $2(0.5)(0.5)(0.012) = 0.006$.
4. Total variance is $0.01 + 0.01 + 0.006 = 0.026$. Volatility is $\sqrt{0.026} = 16.12\%$.

At correlation 1, volatility would remain 20%. At correlation 0, it is 14.14%. Diversification depends on joint movements, not just the number of countries held.

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Estimating Beta from Historical Returns

[Background Context]

A **regression** fits a linear relationship to paired historical returns. Use returns above the risk-free rate in the same currency at each date t:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{M,t} - R_{f,t}) + \varepsilon_{i,t}.$$

- β_i is the fitted slope. α_i is the fitted excess return when market excess return is zero. The residual $\varepsilon_{i,t}$ is the part the fitted line does not explain at date t.
- With covariance 0.024 and market variance 0.020, beta is $0.024/0.020 = 1.20$. With a 3% risk-free rate and 5% market premium, CAPM requires $3\% + 1.2(5\%) = 9\%$.
- CAPM predicts zero expected pricing alpha under its assumptions. A historical estimate can differ because the sample is noisy or the chosen model omits relevant risks.

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A Home-Market CAPM Example

[Background Context]

Home-country benchmark: value a foreign investment from the perspective of a home-market investor when that benchmark is economically justified. It is different from using the project's local market.

1. A Singapore investor evaluates foreign cash flows in SGD. Estimate the project's SGD-return beta relative to a Singapore market portfolio.
2. Inputs: SGD default-free rate 2.5%, home-market equity premium 5%, home-market beta 1.1.
3. Home-benchmark required return is $2.5\% + 1.1(5\%) = 8.0\%$ in SGD. Discount consistently measured SGD expected cash flows.
4. Compare assumptions: a segmented local CAPM uses the project-market benchmark. An integrated global CAPM uses the world portfolio. A multifactor model adds specified priced risks.

Home nationality alone does not establish the correct pricing model. Identify which investors influence the price, their return currency, and their access to other markets. Avoid automatically adding a sovereign spread.

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Stock Connect: The Institutions behind the Trade

1. The broker sends the order through the Hong Kong exchange and the linked Shanghai or Shenzhen exchange. The route depends on the trade direction.

2. Hong Kong Exchanges and Clearing operates the Hong Kong market group. The China Securities Regulatory Commission oversees mainland securities markets.

3. Hong Kong Securities Clearing Company and China Securities Depository and Clearing Corporation (ChinaClear) support clearing and settlement. A nominee holds registered title on behalf of the underlying investor through the custody chain.

4. A trading quota limits permitted flows. Eligibility rules determine which investors and securities can use the link. Market calendars determine when trading and settlement can occur.

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ETF History and Settlement Terms

[Background Context]

- The SPDR S&P 500 ETF launched in the US in 1993. S&P 500 names its large-company US equity index. Active ETFs let managers choose investments under a stated strategy.
- **UCITS** means Undertakings for Collective Investment in Transferable Securities, a European framework for investment funds. The fund's domicile is its legal home.
- **T+1 settlement** means completing delivery one business day after the trade date. Different market calendars and settlement cycles can create timing gaps between ETF units and underlying assets.
- **Assets under management (AUM)** is the value managed by a fund or provider. Size rankings are dated observations, not measures of return or suitability.

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CAPM Reference: Market-Wide and Firm-Specific Risk

- In the model, investors hold the market portfolio. Systematic risk is the market-wide risk that remains after diversification.
- Unsystematic, or firm-specific, risk can be reduced by spreading holdings across companies. CAPM assigns no separate expected-return premium to that diversifiable component.

Security Market Line (returns in one currency)

$$E(R_i) = R_f + \beta_i \left[E(R_m) - R_f\right]$$

- R_f: Risk-free rate.
- $\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}$: Sensitivity of asset i to market portfolio returns.
- $E(R_m) - R_f$: Market risk premium.

CAPM regression and beta calculation →

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Integration → Currency and beta → Numerical case →

Segmented vs. Integrated Capital Markets

[Background Context]

Segmented Capital Markets

- Capital controls and regulatory barriers restrict cross-border capital flows.
- Local supply and demand determine asset prices.
- Investors measure asset risk relative to local market indexes.

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Integrated Capital Markets

- Capital moves freely across national borders without friction.
- Investors require the same compensation for identical risk exposures across countries.
- Investors measure asset risk relative to global market indexes.

CAPM Beta and Currency-Consistent Inputs

For a chosen market portfolio M,

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_M)}{\mathrm{Var}(R_M)}, \qquad E[R_i] = r_f + \beta_i(E[R_M] - r_f).$$

- Beta measures sensitivity to market returns. It depends on correlation and the relative volatility of the asset and market. An asset with beta below one can still have high total volatility from company-specific risk.
- Express the asset, market, and risk-free return in the same currency and over the same horizon. A defaultable sovereign yield is not automatically a risk-free rate.
- Local, home-country, and global benchmarks measure different market risks. Adding a country-risk premium changes the model and requires a reason.

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A Multifactor Model: Exposure Times Risk Premium

A factor is a common source of return variation. The global market is one factor. A model can add specified currency or economic factors if they help explain priced risk.

$$E[R_i] - r_f = \beta_{ig}\lambda_g + \sum_k \beta_{ik}\lambda_k.$$

- i identifies the asset and g the global market. k indexes each additional factor. A beta measures exposure and a lambda measures expected compensation per unit of that exposure.
- All terms use the same return currency and horizon. The sum adds the compensation attributed to each specified exposure.
- Local CAPM uses a local market. Global CAPM uses a global market. This extension adds sources of risk rather than adding a country's entire government-bond yield spread.

Numerical illustration: If $r_f = 3\%$, the market contribution is $1.2(5\%) = 6\%$, and an added factor contributes $0.5(2\%) = 1\%$, the required expected return is 10%.

CAPM Example → | Global Returns → | Home CAPM →

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Choosing an International Asset-Pricing Model

Choose a model based on which risks investors price. Keep every return and discount rate in the same currency.

	Local CAPM	Global CAPM	Multifactor model
Market assumption	Segmented	Integrated	Additional priced risks
Beta benchmark	Local equity market	Global market	Global market and specified factors
Risk premium	Local equity premium	Global equity premium	Premium for each factor
Country risk	Reflected in local inputs	Priced through global covariance	Priced through relevant factor exposures

Use a default-free rate, or a justified proxy, in the cash-flow currency. Add a sovereign spread only if the model gives a reason to do so.

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Appendix: Hedging an Uncertain Foreign Asset Value

[Background Context]

Invest EUR 100 at $S_0 = 8.50$ HKD/EUR, costing HKD 850. The asset earns 5% in EUR. Sell EUR 100 forward at $F = 8.40$ HKD/EUR. At maturity, $S_1 = 7.65$ HKD/EUR.

- Asset value: EUR 105, worth $105(7.65) = 803.25$ HKD. Unhedged return: $803.25/850 - 1 = -5.5\%$.
- Short-forward payoff: $100(F - S_1) = 100(8.40 - 7.65) = 75$ HKD. Combined value: $803.25 + 75 = 878.25$ HKD.
- Hedged return: $878.25/850 - 1 = 3.32\%$. The original EUR 100 is hedged, but the EUR 5 gain remains exposed.
- If EUR 105 were certain in advance, selling that exact amount forward would lock in $105(8.40) = 882$ HKD, or 3.76%. Equity payoffs are uncertain when the hedge is placed.

Ignore default, collateral, taxes, and transaction costs. Forward points make the hedged HKD return differ from the EUR return.

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Appendix: Portfolio Risk Committee Solution

[Background Context]

Portfolio	Mean return	ES (worst 5%)
Best passing: 60% equity, 10% home, full hedge, HKD 5M cash	5.17%	−30.8%
Single-normal choice: 70% equity, 10% home, full hedge, HKD 3M cash	5.37%	−36.7%

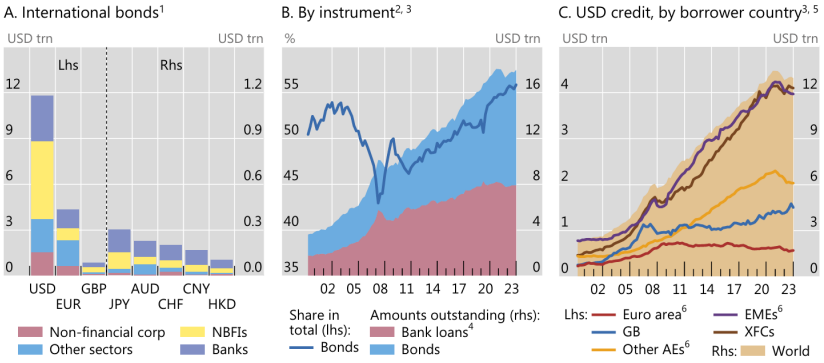
- Both use the Irish fund. The passing portfolio's cash-shortfall probability is 0.3%. The mixture model approves 72 of 243 portfolios. The single-normal model approves 150. Expected ten-year scores are 141.1 and 136.1, respectively.
- Equity-allocation volatility: $20\%\sqrt{0.65} = 16.12\%$; $30\%\sqrt{0.95} = 29.24\%$. Annual fund costs: 0.65%, 0.40%, 0.85%.
- Hedged yield: $1.0\% + (3.5\% - 0.5\%) + 0.5\% = 4.5\%$. Settlement: $20(0.10 - 0.035) =$ HKD 1.3M.
- Cost of selling now: $(3\% - 2.5\%) \times$ HKD 4M = HKD 20,000 for the HK feeder. The Ireland ETF costs HKD 6,000. The US-listed ETF cannot trade. HKD depreciation causes losses for the JPY investor.

ES is a signed return. A more negative value means a larger loss.

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Global Debt Outstanding by Currency

[Background Context]



[1] Amounts outstanding at end-2023. [2] Foreign currency credit denominated in US dollars, euro and yen to borrowers outside the United States, euro area and Japan, respectively. [3] Credit to non-banks, as estimated in the BIS global liquidity indicators. Non-banks comprise non-bank financial entities, non-financial corporations, governments, households and international organisations. [4] Loans by LBS-reporting banks comprising cross-border and local loans. [5] Foreign currency credit denominated in US dollars to borrowers outside the United States. [6] Excluding those in the cross-border financial centres (XFCs) group; see technical annex. Other AEs = other advanced economies.

Sources: Dealogic; Euroclear; LSEG; Xtrakter Ltd; BIS global liquidity indicators; BIS locational banking statistics (LBS).

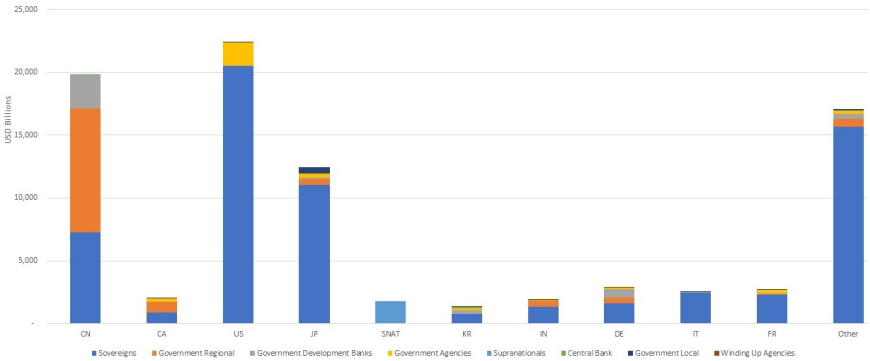
Source: Bank for International Settlements (BIS). USD and EUR dominate international debt issuance.

Amounts outstanding are unpaid debt at a date. NBFIs are non-bank financial institutions. Lhs/rhs mean left/right axes. USD trn means trillions of US dollars.

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Bonds Outstanding: Sovereign & Agency

[Background Context]



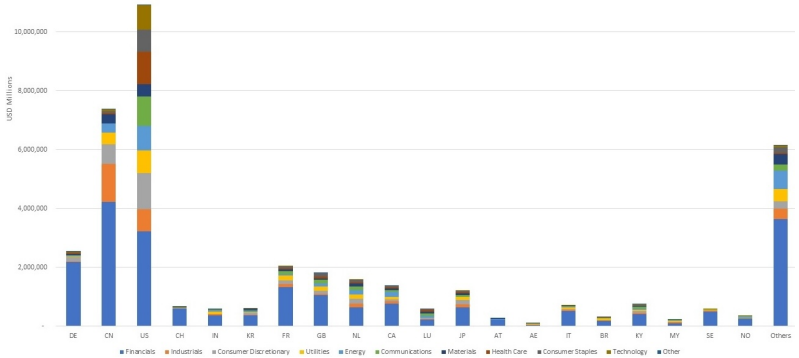
Source: ICMA using Bloomberg Data.

Sovereign means national government. Agencies are government-related issuers. Supranationals are institutions owned by multiple countries. Bars show historical debt outstanding, with currency amounts on the vertical axis.

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Bonds Outstanding: Corporate

[Background Context]



Source: ICMA using Bloomberg Data.

Corporate means company-issued debt. The colors group issuers by business sector. Outstanding amounts are unpaid principal at the observation date, rather than new borrowing during the year.

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Credit Rating Categories

[Background Context]

Category	S&P / Fitch	Moody's
Highest	AAA	Aaa
High	AA	Aa
Upper medium	A	A
Lower medium	BBB	Baa
Speculative	BB	Ba
More speculative	B	B

- S&P, Fitch, and Moody's are examples of rating agencies. Investment grade is the category at or above BBB- / Baa3. Plus/minus signs or numbers are finer rating steps.

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Bond Rating: Categorizing Risk

[Background Context]

Global financial services average cumulative default rates (1981–2022) (%)

	--Time horizon (years)--														
Rating	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**	**Y8**	**Y9**	**Y10**	**Y11**	**Y12**	**Y13**	**Y14**	**Y15**
AAA	0.00	0.04	0.21	0.38	0.55	0.72	0.80	0.89	0.93	0.97	1.02	1.06	1.10	1.15	1.19
AA	0.03	0.09	0.17	0.30	0.44	0.57	0.67	0.78	0.88	0.97	1.04	1.10	1.15	1.21	1.26
A	0.07	0.18	0.31	0.44	0.57	0.71	0.86	0.97	1.06	1.17	1.26	1.33	1.40	1.47	1.56
BBB	0.21	0.55	0.92	1.29	1.60	1.88	2.12	2.35	2.61	2.82	3.07	3.33	3.60	3.90	4.19
BB	0.59	1.45	2.35	3.17	3.93	4.68	5.24	5.61	5.93	6.20	6.54	6.83	6.92	6.96	7.02
B	2.37	4.84	6.77	8.14	9.27	10.29	10.98	11.71	12.46	13.19	13.93	14.44	14.86	15.32	15.66
CCC/C	16.89	21.85	24.58	26.68	27.97	28.63	29.56	30.77	31.29	31.87	31.87	32.20	32.89	33.66	34.08
Investment grade	0.10	0.25	0.44	0.64	0.82	1.00	1.15	1.29	1.42	1.55	1.67	1.77	1.89	2.00	2.12
Speculative grade	2.17	3.90	5.31	6.43	7.37	8.21	8.85	9.41	9.91	10.37	10.84	11.21	11.46	11.71	11.89
All rated	0.48	0.92	1.32	1.68	1.99	2.27	2.51	2.71	2.90	3.07	3.24	3.38	3.51	3.64	3.77

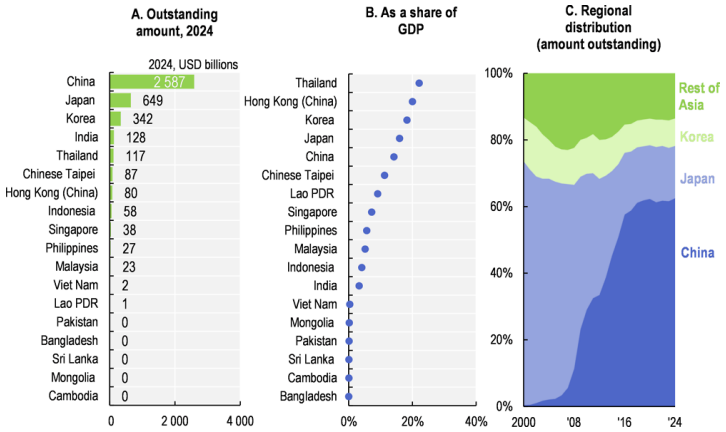
Includes the bank, nonbank financial institution, and insurance sectors. Sources: S&P Global Ratings Credit Research & Insights and S&P Global Market Intelligence's CreditPro®.

Source: Rutherford Rede. Each cell is the historical percentage of issuers in an initial rating group that defaulted by the stated horizon. NR = not rated.

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Corporate Bonds in Asia

[Background Context]

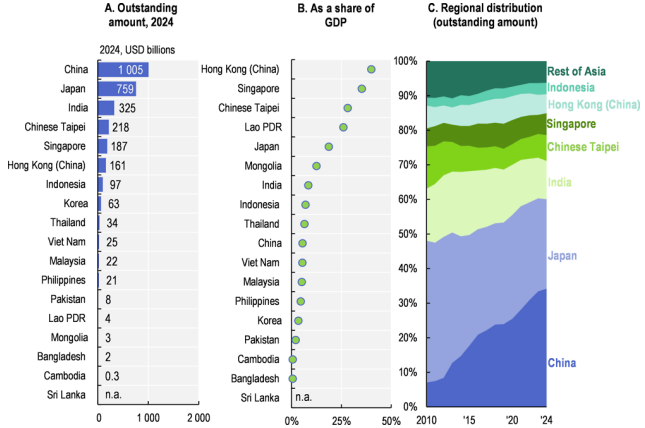


GDP is gross domestic product, the value of final production over a period. Outstanding debt is a stock. Dividing debt by GDP scales market size relative to an economy. A regional share instead divides by all covered Asian debt.

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Syndicated Loans in Asia

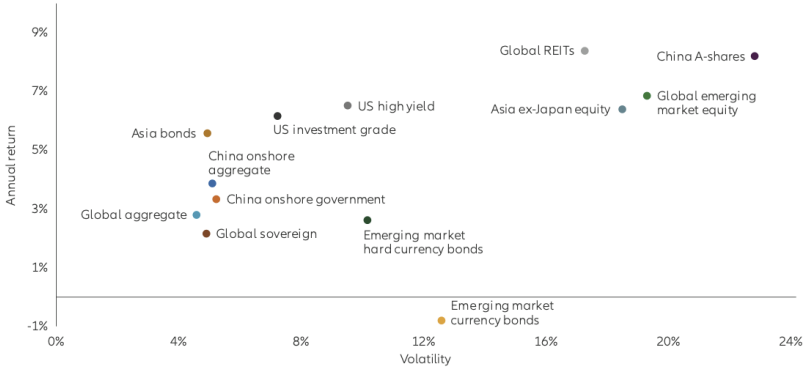
[Background Context]



A syndicated loan has several participating lenders. The charts compare unpaid loan amounts, amounts divided by GDP, and shares of the Asian total. They cover this subset of lending.

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Exhibit 8: 3-year risk/return profile (in USD)



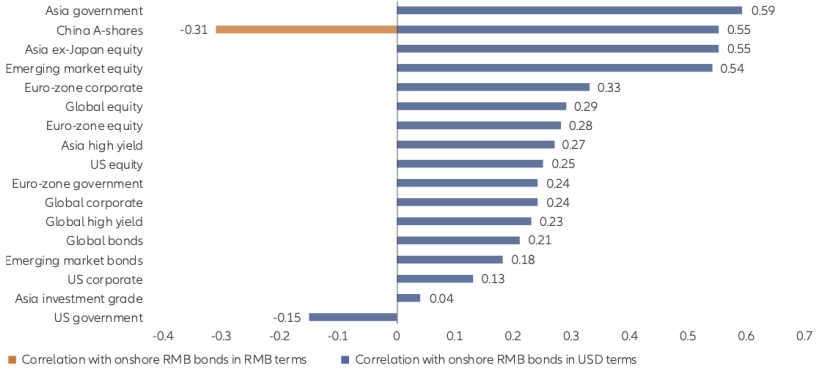
Source: Bloomberg, Allianz Global Investors. Data as at 31 March 2021. Please see the disclosure at the end of this document for important index information.

Historical annual returns and volatility in USD, sample ending March 2021. A REIT is a real-estate investment trust. High yield refers to lower-rated debt. Hard-currency debt is denominated in a major external currency such as USD.

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Chinese Corporate Bonds: Realized Risk and Return

Exhibit 7: 3-year return correlation of onshore RMB bonds vs other major asset classes



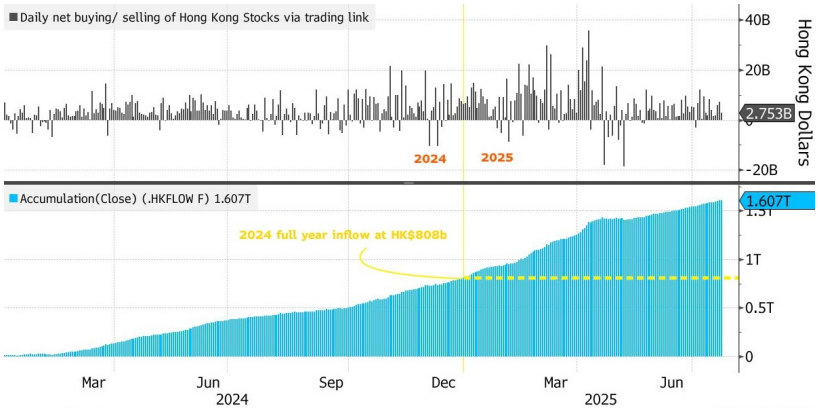
Source: Bloomberg. Data as at 31 December 2020.

Historical return correlations through December 2020. RMB is renminbi. Onshore bonds trade in the mainland market. Comparing RMB and USD returns shows how currency conversion changes measured co-movement.

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China Cross-Border Equity Links: Stock Connect

[Background Context]

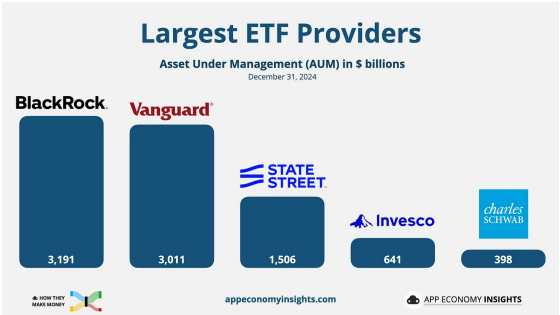


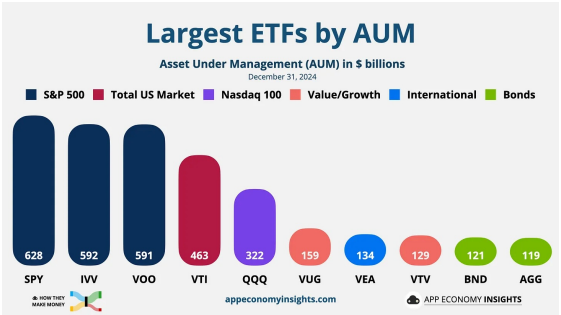
Net buying subtracts sales from purchases. The upper panel shows daily net buying, while the lower panel accumulates it over time. HKD B means billions and T trillions. These flow measures do not measure investment returns.

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Facts about ETFs

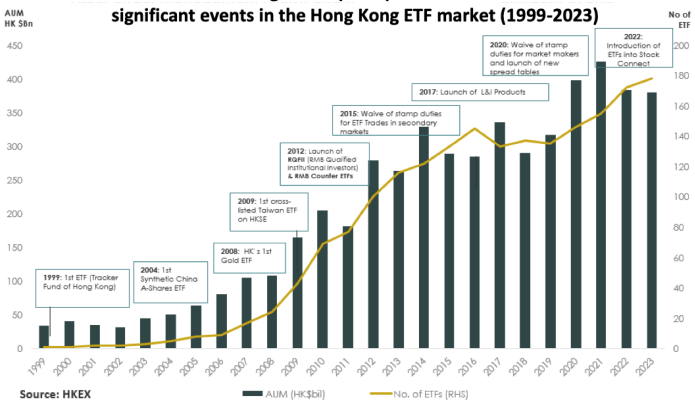
[Background Context]





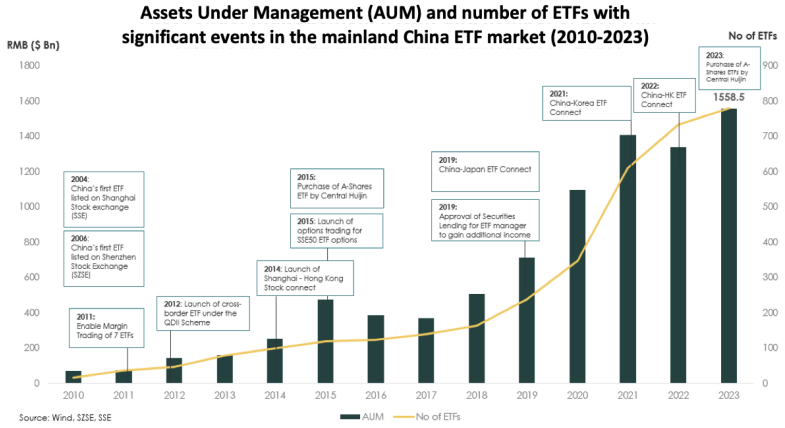
December 2024 AUM rankings. AUM means assets under management. Names at left identify fund providers, and short codes at right identify traded funds (tickers). Dollar amounts measure managed assets, not returns.

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Historical Hong Kong ETFs through 2023. AUM is assets under management. ETP means exchange-traded product, a broader category than ETF. Bars show managed asset value. The line counts products on its separate axis.

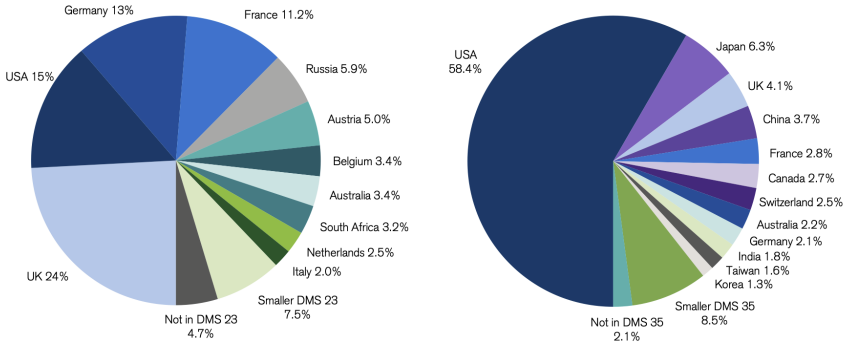
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Historical mainland ETFs through 2023. RMB B means billions of renminbi. AUM is assets under management. Compare the asset-value bars and product-count line using their separate axes.

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Figure 2: Relative sizes of world stock markets, end-1899 (left) versus start-2023 (right)

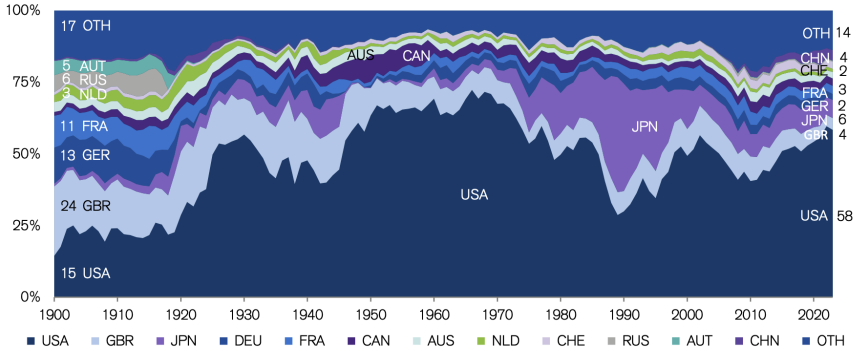


Sources: Elroy Dimson, Paul Marsh and Mike Staunton, DMS Database 2023, Morningstar; data for the right-hand chart from FTSE Russell All-World Index Series Monthly Review, December 2022. Not to be reproduced without express written permission from the authors.

Each slice is a market's share of the covered global equity value at that date. The panels compare end-1899 and start-2023. DMS names the authors' historical database. Countries outside its detailed sample form a separate group.

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Figure 3: The evolution of equity markets over time from end-1899 to start-2023



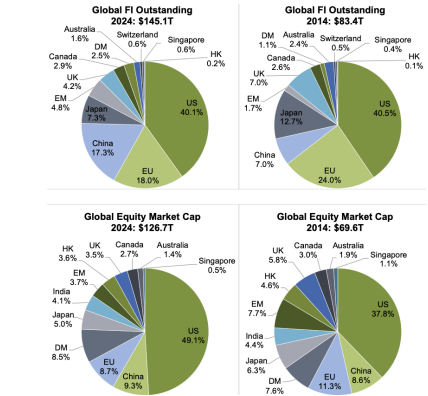
Sources: Elroy Dimson, Paul Marsh and Mike Staunton, DMS Database 2023, Morningstar, and FTSE Russell All-World Index Series weights (recent years). Not to be reproduced without express written permission from the authors.

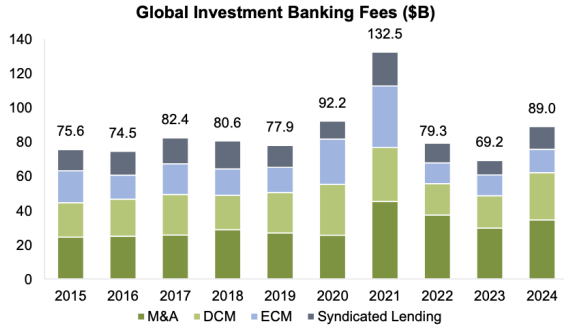
Each colored area is a country's share of covered global equity market value. All shares sum to 100% at each date. Issuance, price changes, and currency conversion can change a country's weight.

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Debt, Equity, and Investment-Banking Fees

[Background Context]





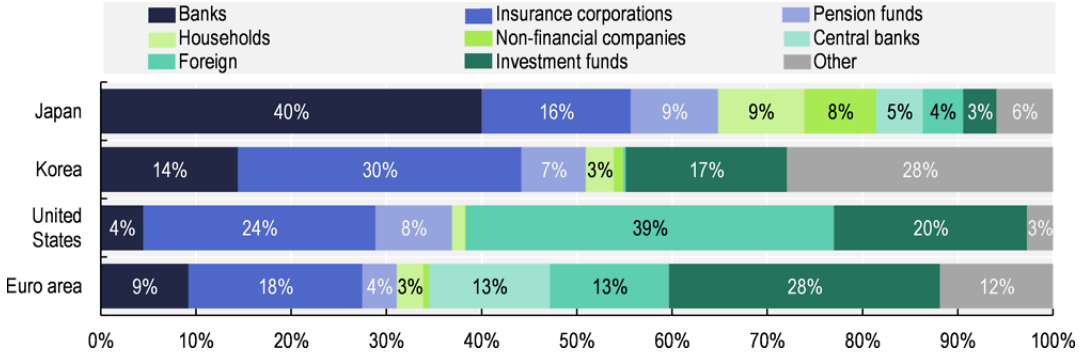
Source: SIFMA (2025).

FI means fixed income (debt securities). M&A is mergers and acquisitions. DCM and ECM are debt- and equity-capital-market services. The left panels show outstanding market size. The right bars show annual banking fees.

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Investor Composition of Bonds

[Background Context]



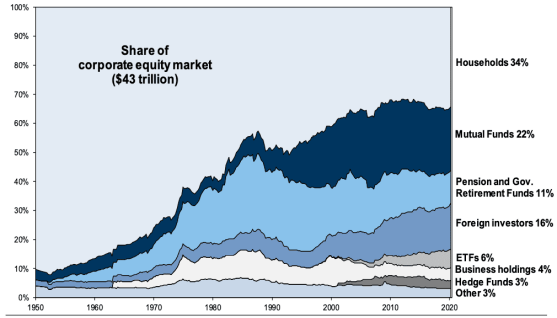
Pension funds invest for retirement payments. Insurers invest against policy claims. Investment funds pool client assets. Each bar divides a bond market's holdings across investor groups, so its shares sum to 100%.

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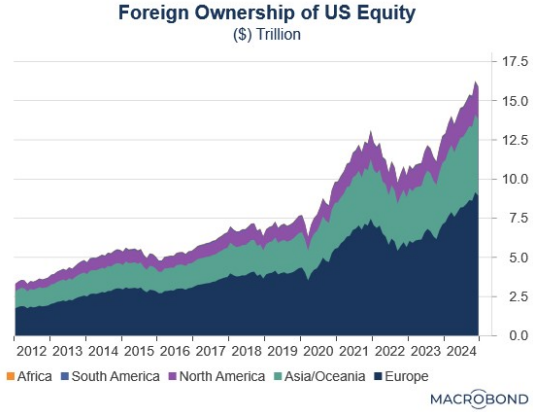
Investor Composition of US Equities

[Background Context]

Exhibit 1: Ownership breakdown of the US equity market
as of March 31, 2020



Source: Federal Reserve Board and Goldman Sachs Global Investment Research



Mutual funds combine investors' money. Retirement funds invest for pension payments. The left panel shows ownership shares. The right panel shows dollar values held abroad, which can rise through price changes as well as purchases.

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Country Diversification and Economic Exposure

[Background Context]

- Country classifications describe one dimension of a portfolio. Holdings can also overlap in sectors, large companies, and sensitivity to global demand.
- A global fund plus a technology fund can increase exposure to companies already among the global fund's largest holdings.
- A firm's listing currency and country do not fully describe its operating exposure. Revenues, production costs, and financing may span several currencies.
- A useful holdings review reports country weights, sector weights, and the largest combined positions across funds.

Example: A company is 5% of a global fund and 10% of a technology fund. A 70/30 combination has $0.70(5\%) + 0.30(10\%) = 6.5\%$ in that company.

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Portfolio Weights and Estimation Error

[Background Context]

- **Mean-variance optimization** chooses weights by balancing expected return against return variance. Estimates of means, volatilities, and correlations depend on the data sample.
- An optimizer can put large weights on a market with a slightly higher estimated return. Data from a different period can reverse the ranking.
- Repeat the calculation with lower expected returns and higher correlations. Large changes in the chosen weights show sensitivity to uncertain inputs.
- Benchmark weights and investment limits can make a portfolio easier to maintain. Limits can restrict individual holdings or turnover. Turnover is the amount traded relative to portfolio size. Give a reason for each limit.

Question: Your model assigns the largest weight to last year's best-performing country. What evidence suggests that its future return will be higher?

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Small Contributions and Fixed Trading Costs

[Background Context]

Illustration: HKD 3,000 monthly contribution. Each purchase costs HKD 30 plus a 0.25% FX conversion charge. Ignore other charges.

Purchase size	Fixed fee	FX charge	Cost relative to purchase
HKD 3,000	30	7.50	1.25%
HKD 9,000	30	22.50	0.58%

- Splitting HKD 3,000 across three orders would triple the assumed fixed fee. Small accounts may benefit from fewer holdings or directing contributions to one underweight holding.
- Combining purchases lowers fixed costs but leaves cash uninvested for longer. Compare available brokers and recurring-investment arrangements using actual fee schedules.

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ETF Execution across Time Zones

[Background Context]

- Review: Check the live bid and ask. A limit buy order caps the purchase price, but may remain unfilled. A market order leaves the execution price uncertain.
- When underlying markets are closed, dealers may face more hedging uncertainty. Quoted spreads and apparent premiums can widen.
- A published NAV can use older asset prices. Compare the current ETF price with current asset values and trading costs before calculating an arbitrage profit.
- Before a trade, check the correct share class, trading counter, order size, and available settlement currency. Review the execution price and fees afterwards.

Source: IFEC, ETF major risks.

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Distributions, Domicile, and Total Return

[Background Context]

- An accumulating share class retains income. A distributing class pays cash that the investor can spend or reinvest.
- **Illustration**: a unit worth 100 pays a distribution of 4. If its value falls to 96 solely because of the payout, total wealth remains $96 + 4 = 100$ before taxes and costs.
- Fund domicile determines the legal fund jurisdiction.
- Investor residence and fund structure can affect taxes and access.

Source: iShares, How to Compare Global ETFs.

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Illustration: a 60/40 policy portfolio is compared with a 60% global-equity and 40% defensive-asset benchmark. All returns are HKD total returns over the same year.

$$R_{\text{benchmark}} = 0.60(10\%) + 0.40(2\%) = 6.8\%.$$

- Suppose the portfolio earns 6.1% net of costs. This return is 0.7 percentage points below the benchmark. Comparing the whole portfolio only with US equities answers a different question.
- Attribute the difference to allocation, fund returns, currency policy, and costs. One year's difference is weak evidence of investment skill.
- A time-weighted return combines returns between deposits and withdrawals. A money-weighted return equates the present values of deposits with withdrawals and final value. It depends on when the investor adds or removes cash.

The arithmetic assumes initial benchmark weights held through this year and no interim rebalancing.

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Example policy statement:

- "I separate near-term spending from long-term investments and measure each goal in its spending currency."
- "My long-term target is 60% global equities and 40% defensive assets. I understand the losses illustrated in the stress test."
- "I use a stated global benchmark for equities. I calculate the combined holdings of my funds and document any regional or sector overweight."
- "I specify the currency policy and compare total costs. I direct contributions to underweight assets and apply my chosen rebalancing rule."

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Swap Dealer Regulation: US and Hong Kong Examples

[Background Context]

- **United States**: The SEC generally regulates swaps on one stock or a narrow stock index. The CFTC generally regulates broad equity-index and interest-rate swaps. Dealer registration depends on definitions and thresholds.
- **Entity matters**: US bank dealers follow their banking regulators' capital and margin rules. SEC rules for nonbank dealers vary with entity type, collateral requirements, and permitted exceptions.
- **Hong Kong banks**: HKMA CR-G-14 covers margin and risk controls for derivatives traded outside exchanges without central clearing. Its application depends on counterparties, products, and thresholds. Controls include written terms, valuations, record comparisons, and dispute resolution.
- **Product matters**: Mandatory clearing and trading rules apply to specified products and participants. Check the product and parties to determine clearing duties and margin requirements.

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Source: SEC, security-based swap framework; SEC, nonbank capital/margin rules; HKMA CR-G-14. Checked 28 October 2026.